

Prague University of Economics and Business

Faculty of Informatics and Statistics



**AUTONOMOUS VEHICLES AND RISK MANAGEMENT FOR
CRITICAL TRAFFIC SITUATIONS**

BACHELOR THESIS

Study programme: Applied Informatics

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Abstract

This thesis aims to describe, assess, and manage the risks for critical traffic situations associated with the deployment of autonomous vehicles (AVs), touching on the application of insights from Information and Communication Technology (ICT) process and risk management. The study will explore the ethics of autonomous vehicles to evaluate the impact of traffic collision situations. Furthermore, quantitative research will be conducted to investigate respondents' attitudes towards the risk of the described traffic situations associated with AV deployment.

Keywords

Artificial Intelligence, Autonomous driving, critical traffic situations, risk management, ethics, traffic collision situations

JEL Classification

M15

Contents

Contents.....	4
1 Introduction	7
1.1 Background and Motivation	7
1.2 Objectives and Scope of the Study	8
1.3 Structure of the Thesis	9
2 Literature Review and Typical Traffic Collision Situations	10
2.1 Search Strategy and Selection Criteria	10
2.2 Description of Typical Traffic Collision Situations	10
2.3 Pedestrian and Cyclist Accidents	12
2.4 Impact of AV Deployment on Traffic Situations	13
2.5 Potential Benefits	15
2.6 Potential Risks and Challenges	16
3 Autonomous Vehicles, Ethics, and ICT Process Management	19
3.1 Autonomous Vehicle Technology	19
3.2 Ethics in Autonomous Vehicles	25
3.3 Ethical Theories and Approaches	27
4 Process Maps and Risk Matrices	33
4.1 Selection of Traffic Situations for Process Mapping	34
4.2 Detailed Process Maps for Selected Situations	36
4.3 Risk Matrix	36
4.4 Creation of Risk Matrix	38
4.5 Risk Assessment Criteria	40
4.6 Risk Matrix Development and Analysis	42
5 Quantitative Research	45
5.1 Survey Design	45
5.2 Sampling Strategy and Population	48
5.3 Statistical Analysis and Software Tools	50
5.4 Interpretation and Implications of Results	52
6 Evaluation and Conclusion	57
6.1 Theoretical and Practical Implications	57
6.2 Discussion of Findings and Conclusions	58

6.3 Limitations of the Study	59
6.4 Recommendations for Future Research	60
7 List of references.....	62

List of Abbreviations

BT	Bachelor Thesis
FIS	Faculty of Informatics and Statistics
AV	Autonomous Vehicle
AI	Artificial Intelligence

1 Introduction

1.1 Background and Motivation

With over two years of experience as a working student at Porsche Engineering, my interest in autonomous vehicles and their potential influence on society has significantly expanded. My perspective was also greatly influenced by the importance of automotive standards, which I mention later in this thesis.

Technological progress and artificial intelligence have enabled the creation of these vehicles, enabling new opportunities for enhanced safety, efficiency, and convenience in transportation. This experience in the automotive industry has granted me a deeper understanding of the challenges encountered by engineers, designers, and manufacturers in building such systems while also focusing my interest on the ethical considerations and risk management aspects of autonomous vehicles.

Ultimately, this thesis seeks to provide a comprehensive understanding of the various factors at play when it comes to autonomous vehicles, risk management, and ethical decision-making in critical traffic situations. By doing so, it aims to contribute to the development of safer, more reliable, and ethically responsible autonomous transportation systems for the future.

The emergence of autonomous vehicles (AVs) promises to revolutionize transportation by enhancing safety and efficiency. However, integrating AVs into existing infrastructure presents challenges, particularly in managing critical traffic situations. Examining risk management strategies is crucial for ensuring safety and maintaining public trust. Ethical implications are central to AV development, as decisions in critical situations involve complex dilemmas involving stakeholders, legal frameworks, and societal values. The application of Information and Communication Technology (ICT) process and risk management can significantly contribute to understanding and addressing risks in these traffic situations.

1.2 Objectives and Scope of the Study

Research question

What are the key differences in opinions and expectations between AI/automotive engineers and the general public regarding autonomous vehicle decision-making in critical traffic situations, and which group expresses more trust towards AVs?

Overview

The main objectives of this study are to explore the risks associated with critical traffic situations involving autonomous vehicles, understand the ethical implications of AV deployment, and develop strategies for managing these risks through an ICT process and risk management approach. Specifically, the study aims to:

1. **Provide an overview of autonomous vehicle technology and its ethical implications:** This objective focuses on understanding the current state of AV technology, its potential benefits, and the ethical challenges it presents. The study will investigate the moral dilemmas and ethical principles relevant to the development and deployment of AVs.
2. **Explore the relationship between ICT process management and risk management in the context of AV deployment:** This objective seeks to examine the application of ICT process management principles and methodologies to the risk management process for AVs. By analyzing the parallels between these fields, the study aims to identify best practices and approaches for addressing risks in critical traffic situations involving AVs.
3. **Identify and analyze typical traffic collision situations that may arise from the deployment of AVs:** The study aims to review existing literature to identify and categorize common traffic collision scenarios that could occur with the integration of AVs into the transportation infrastructure. Understanding these scenarios will help in the development of risk management strategies for each situation.
4. **Design and conduct quantitative research to investigate respondents' attitudes towards the risks associated with critical traffic situations involving AVs:** The study aims to gather public opinions and perceptions regarding the risks of AV deployment in critical traffic situations. Quantitative research methods, such as surveys or questionnaires, will be employed to collect data from respondents. This information will provide insights into the public's level of trust and acceptance of AVs in various traffic scenarios.
5. **Evaluate the theoretical and practical findings of the study and provide recommendations for future research:** The final objective is to analyze the study's findings and evaluate their implications for both the theoretical and practical aspects of AV risk management. The study will also

identify limitations and suggest areas for further research to address the challenges and ethical concerns related to AV deployment in critical traffic situations.

1.3 Structure of the Thesis

The structure of this thesis is organized into two main sections, focusing on both theoretical and practical aspects of autonomous vehicles. The first section delves into the theoretical foundation, starting with the history of autonomous vehicles, followed by a literature review. This section further explores various traffic situations and the role of process management in the context of autonomous vehicles. The second section concentrates on the practical application of the research, beginning with the design and execution of a survey. It then discusses the outcomes of the survey, leading to the conclusion of the study, which includes a synthesis of the findings and their implications. Finally, the thesis offers recommendations for further research and potential improvements in the field of autonomous vehicles and risk management.

2 Literature Review and Typical Traffic Collision Situations

2.1 Search Strategy and Selection Criteria

Systematic literature review methodology:

My literature research followed the PRISMA scheme – records identified (both through database searching and other sources) $n = 17$. I removed 6 sources through a screening process. The full text articles assessed for eligibility were . The number of resources included in the synthesis and referred to in the thesis is 11.

2.2 Description of Typical Traffic Collision Situations

Overview

This chapter describes the typical traffic collision situations, drawing inspiration from the MIT Moral Machine experiment. This thesis aims to understand the ethical considerations and potential decision-making processes of autonomous vehicles in various critical traffic scenarios. By examining these situations, we can gain valuable insights into the challenges and complexities that AVs may encounter in real-world driving conditions. The analysis of these scenarios will provide a foundation for further discussion on the ethical implications, process management, and risk mitigation strategies for autonomous vehicles in the subsequent chapters of this thesis.

Traffic Collision Types

- *Single-vehicle Accidents:* Crashes involving only one vehicle, which may be caused by factors such as loss of control, driving off the roadway, or colliding with a stationary object.

Analysis of Traffic Collision Situations for Autonomous Vehicles

- *Sensor and Perception Failures:* Failures in sensing and perception systems (e.g., lidar, radar, cameras) could lead to collisions, such as the inability to detect an obstacle, misclassifying an object, or inaccurately estimating vehicle speed or distance.

- *Decision-making and Control Errors*: Errors in the decision-making and control algorithms of autonomous vehicles could contribute to collisions, including improper braking, acceleration, or steering, as well as misinterpretation of traffic signals or signs.
- *Human-Autonomous Vehicle Interaction*: Traffic collision situations that may arise from the interaction between human-driven vehicles and autonomous vehicles, such as humans misinterpreting the intentions of an autonomous vehicle or autonomous vehicles not being able to predict human behavior accurately.
- *Infrastructure and Environmental Factors*: Role of road conditions, weather, and infrastructure in contributing to traffic collision situations involving autonomous vehicles, including challenges such as poor visibility, road debris, or deteriorating pavement markings.

Potential Collision Scenarios in Mixed Traffic

- *Scenario 1*: An autonomous vehicle fails to recognize a stop sign obscured by foliage, leading to a side-impact collision with a human-driven vehicle at an intersection.



(Slight Street Sign Modifications Can Completely Fool Machine Learning Algorithms - IEEE Spectrum, n.d.)

- *Scenario 2*: An autonomous vehicle experiences a software glitch that causes it to unexpectedly brake, resulting in a rear-end collision with a human-driven vehicle following too closely.
- *Scenario 3*: A human-driven vehicle attempts an unsafe lane change, sideswiping an autonomous vehicle traveling in the adjacent lane.
- *Scenario 4*: An autonomous vehicle misinterprets a traffic signal and proceeds through an intersection, colliding with a human-driven vehicle making a left turn. (Watzenig & Horn, 2017)

2.3 Pedestrian and Cyclist Accidents

Pedestrian and Cyclist Accidents Involving Autonomous Vehicles

- *AV perception and decision-making errors:* How autonomous vehicles' sensing and decision-making systems might contribute to pedestrian and cyclist accidents, such as failing to detect vulnerable road users in time, misjudging their speed or trajectory, or not accounting for unpredictable behaviors.
- *Human-AV miscommunication:* Potential accidents involving pedestrians and cyclists that may result from miscommunication between human road users and autonomous vehicles, including ambiguous signaling, misinterpretation of AV behavior, or overreliance on AV technology.
- *AV system limitations:* The role of current limitations in autonomous vehicle systems, such as the inability to recognize hand signals from cyclists or pedestrians, or difficulty handling low-visibility situations where vulnerable road users are harder to detect.

Strategies for Reducing Pedestrian and Cyclist Accidents in AV Deployment

- *Enhanced sensor systems and algorithms:* Developing improved sensors and perception algorithms to better detect and predict the actions of pedestrians and cyclists, helping autonomous vehicles make safer and more informed decisions.
- *Vehicle-to-pedestrian (V2P) and vehicle-to-cyclist (V2C) communication:* Implementing V2P and V2C communication systems to facilitate the exchange of real-time traffic information between vehicles and vulnerable road users, reducing the risk of accidents by enabling better coordination and decision-making.
- *Improved AV-human interaction:* Designing better communication methods for autonomous vehicles to signal their intentions to pedestrians and cyclists, helping to reduce misunderstandings and misinterpretations that may lead to accidents.
- *Vulnerable road user awareness and education:* Promoting awareness and education among pedestrians and cyclists regarding the capabilities and limitations of autonomous vehicles, encouraging safer and more cooperative behaviors around AVs.
- *Infrastructure improvements:* Collaborating with transportation agencies to optimize road designs and traffic control systems to accommodate the unique needs and capabilities of autonomous vehicles and enhance the safety

of vulnerable road users, such as creating dedicated bike lanes, enhancing crosswalk visibility, or implementing pedestrian and cyclist detection systems at intersections.(Watzenig & Horn, 2017)

2.4 Impact of AV Deployment on Traffic Situations

Overview

In this chapter I explore the impact of autonomous vehicle (AV) deployment on various traffic situations. As AVs are becoming increasingly integrated into the transportation ecosystem, it is crucial to understand their potential effects on road safety, traffic flow, and infrastructure. This chapter discusses the transformative nature of AVs and their potential to reshape transportation patterns, while also considering the possible challenges and concerns that may arise. We will examine how AVs can influence driver behavior, vehicle interactions, and the overall transportation system, and investigate the implications of these changes for traffic management and policy development.

Potential Benefits of AV Deployment

- *Improved traffic efficiency:* Autonomous vehicles can communicate with each other and coordinate their movements, potentially reducing traffic congestion, optimizing traffic flow, and shortening travel times.
- *Enhanced road safety:* AVs are designed to obey traffic laws and adhere to safe driving practices, which may contribute to a reduction in traffic accidents caused by human error, such as speeding, distracted driving, or impaired driving.
- *Increased accessibility:* AVs can provide mobility solutions for individuals with disabilities or those who are unable to drive, increasing access to transportation and reducing social isolation.
- *Reduced environmental impact:* The deployment of autonomous vehicles can potentially result in more efficient driving, optimized routing, and increased adoption of electric vehicles, leading to reduced emissions and a lower environmental impact.(Awad, 2017)

Challenges and Concerns in AV Deployment

- *Mixed traffic:* During the transition period, autonomous vehicles will share the roads with human-driven vehicles, which could create challenges in

communication and interaction, potentially leading to accidents or disruptions in traffic flow.

- *Cybersecurity risks:* As autonomous vehicles rely on software systems and networked communications, they are susceptible to cybersecurity attacks, which could compromise safety and create potentially dangerous traffic situations.

- *Regulatory and legal considerations:* The development and deployment of AVs require new regulations and legal frameworks to address issues such as liability, insurance, data privacy, and safety standards, which could impact their adoption and integration into traffic systems.(Awad, 2017)

Impact of AV Deployment on Critical Traffic Situations

- *Alleviating human error:* Autonomous vehicles can potentially reduce the occurrence of critical traffic situations caused by human error, such as intersection accidents, lane change and merging accidents, and rear-end collisions, by adhering to safe driving practices and using advanced sensing and decision-making systems.

- *Adapting to unpredictable situations:* AVs may struggle to handle critical traffic situations that involve unpredictable behaviors from human drivers, pedestrians, or cyclists, potentially leading to accidents or disruptions in traffic flow.

- *Transitioning between autonomous and manual control:* Situations where an autonomous vehicle must hand over control to a human driver in a critical traffic situation can present challenges in communication, reaction time, and decision-making, potentially increasing the risk of accidents.

- *Infrastructure adaptations:* The impact of AV deployment on critical traffic situations may be influenced by the extent to which existing transportation infrastructure is updated and adapted to accommodate autonomous vehicles, such as dedicated AV lanes, intelligent traffic management systems, or improved road markings and signage.(Watzenig & Horn, 2017)

Strategies to Address the Impact of AV Deployment on Critical Traffic Situations

- *Developing robust AV systems:* Ensuring that autonomous vehicles are equipped with advanced sensors, algorithms, and communication systems to handle a wide range of traffic situations, minimize errors, and enhance safety.

- *Promoting collaboration between stakeholders:* Encouraging cooperation among policymakers, transportation agencies, AV manufacturers, and other stakeholders to address challenges and develop comprehensive strategies for integrating autonomous vehicles into traffic systems.
- *Educating the public:* Increasing awareness and understanding of autonomous vehicle technology and its limitations among the general public, and encouraging safe and responsible interactions between human road users and AVs.
- *Continuous monitoring and improvement:* Implementing ongoing monitoring and evaluation of the impact of AV deployment on critical traffic situations to identify areas for improvement, inform future research and development, and ensure that autonomous vehicles contribute to safer and more efficient transportation systems.(Watzenig & Horn, 2017)

2.5 Potential Benefits

Improved traffic efficiency

- *Coordinated traffic flow:* AVs can communicate and coordinate their movements, which allows them to maintain optimal speeds, minimize unnecessary stops, and respond more effectively to changing traffic conditions.
- *Reduced traffic congestion:* As autonomous vehicles can maintain consistent speeds and follow each other more closely, they can potentially reduce the number and severity of traffic jams, resulting in more efficient use of existing road infrastructure.
- *Optimized routing:* AVs can use real-time traffic data to choose the most efficient routes and avoid congested areas, reducing overall travel times and improving the utilization of transportation networks.(Watzenig & Horn, 2017)

Enhanced road safety

- *Reduced human error:* AVs are programmed to obey traffic laws and can maintain constant vigilance, potentially reducing the number of accidents caused by human errors such as distraction, fatigue, or impaired driving.
- *Advanced collision avoidance:* Autonomous vehicles employ advanced sensor systems and artificial intelligence to predict and react to potential collisions, which can help to reduce the occurrence of accidents in various traffic situations.

- *Predictable driving behavior*: AVs are designed to drive consistently and predictably, making it easier for other road users to anticipate their actions and potentially reducing the risk of collisions.(Koopman & Wagner, 2017)

Increased accessibility

- *Mobility for all*: Autonomous vehicles can provide a means of transportation for individuals who are unable to drive due to disabilities, age, or other limitations, increasing their access to essential services, employment opportunities, and social connections.
- *Reduced reliance on personal vehicles*: The deployment of AVs in public transportation and shared mobility services can potentially reduce the need for private car ownership, making transportation more accessible and affordable for a wider range of users.

Reduced environmental impact

- *Energy-efficient driving*: Autonomous vehicles can optimize their driving behavior to minimize energy consumption, such as maintaining consistent speeds, using regenerative braking, and avoiding unnecessary idling, which can lead to reduced fuel consumption and lower emissions.
- *Integration with electric vehicles*: The deployment of AVs can potentially encourage the adoption of electric vehicles, which produce fewer emissions and have a lower environmental impact than traditional internal combustion engine vehicles.
- *Smart infrastructure integration*: AVs can interact with intelligent transportation systems and infrastructure to optimize traffic flow, reducing the overall environmental impact of transportation networks by minimizing congestion and idling.(Watzenig & Horn, 2017)

2.6 Potential Risks and Challenges

Mixed traffic

- *Communication challenges*: Autonomous vehicles and human-driven vehicles may face difficulties in effectively communicating their intentions to each other, potentially leading to misunderstandings and accidents.
- *Inconsistent driving behavior*: The interaction between AVs, which exhibit predictable and rule-abiding behavior, and human-driven vehicles, which

can exhibit unpredictable or aggressive driving, may create challenges in maintaining smooth traffic flow and avoiding collisions.

Cybersecurity risks

- *Vulnerability to attacks:* Autonomous vehicles rely on software systems and networked communications, making them susceptible to hacking, malware, or other forms of cybersecurity attacks that could compromise vehicle safety and create hazardous traffic situations.
- *Data privacy concerns:* The collection and transmission of data by autonomous vehicles, such as location information or user preferences, can raise privacy concerns and require robust data protection measures to prevent unauthorized access or misuse of sensitive information.

Regulatory and legal considerations

- *Liability and insurance:* The deployment of autonomous vehicles introduces new questions regarding liability in the event of an accident, requiring updates to existing legal frameworks and insurance policies to address the unique characteristics of AVs.
- *Safety standards:* Regulators and lawmakers must establish safety standards and performance requirements for autonomous vehicles to ensure their safe integration into traffic systems, which may pose challenges in keeping pace with rapidly evolving AV technology.

Source	Cause of hazardous event	Within scope of
System	E/E System failures	ISO 26262 series (Functional Safety)
	Performance limitations or insufficient situational awareness, with or without reasonably foreseeable misuse	ISO/DIS 21448* (SOTIF)
	Reasonably foreseeable misuse, incorrect HMI (e.g. user confusion, user overload)	ISO/DIS 21448* ISO 26262 series European statement of principal on the design of human-machine-interface
	Hazards caused by the system technology	Specific standards
External factor	successful attack exploiting vehicle security vulnerabilities	ISO 21434** or SAE J3061
	Impact from active Infrastructure and/or vehicle to vehicle communication, external devices and cloud services.	ISO 20077 series ISO 26262 series
	Impact from car surroundings (other users, "passive" infrastructure, environmental conditions: weather, Electro-Magnetic Interference, etc.)	ISO/DIS 21448* ISO 26262 series

* ISO/DIS 21448:2019 – Draft International Standard version

** ISO/SAE DIS 21434 - Draft International Standard version

(Application of Functional Safety in Autonomous Vehicles Using ISO 26262 Standard: A Survey | IEEE Conference Publication | IEEE Xplore, n.d.)

- *Ethical considerations:* Policymakers need to address the ethical implications of AV deployment, such as how autonomous vehicles should

prioritize safety in complex or morally ambiguous situations, which may require ongoing public dialogue and stakeholder input.(Watzenig & Horn, 2017)

Technological limitations

- *Perception and decision-making errors:* Autonomous vehicles may struggle to handle certain traffic situations or environments, such as extreme weather conditions, poor visibility, or unstructured environments, which could lead to errors in perception or decision-making and increase the risk of accidents.
- *Infrastructure compatibility:* The successful deployment of AVs depends on their ability to operate within existing transportation infrastructure, which may require upgrades or modifications to accommodate the unique needs of autonomous vehicles, such as improved road markings, signage, or dedicated AV lanes.
- *Transition between control modes:* The handover of control between an autonomous vehicle and a human driver may present challenges, such as delays in reaction time or miscommunication, which could increase the risk of accidents during critical traffic situations.(Awad, 2017)

3 Autonomous Vehicles, Ethics, and ICT Process Management

3.1 Autonomous Vehicle Technology

Overview

This chapter explains the fundamental components that make up autonomous vehicles, including the history, hardware, software, AI , and supporting infrastructure. By examining these components, the thesis sets the stage for a deeper exploration of the ethical considerations and risk management strategies in the subsequent chapters.

History and Development of Autonomous Vehicles

- *Early Concepts and Visions:* Discussion of the initial ideas and visions of self-driving vehicles, including mentions of futurist predictions and science fiction portrayals.(Maurer et al., 2016)

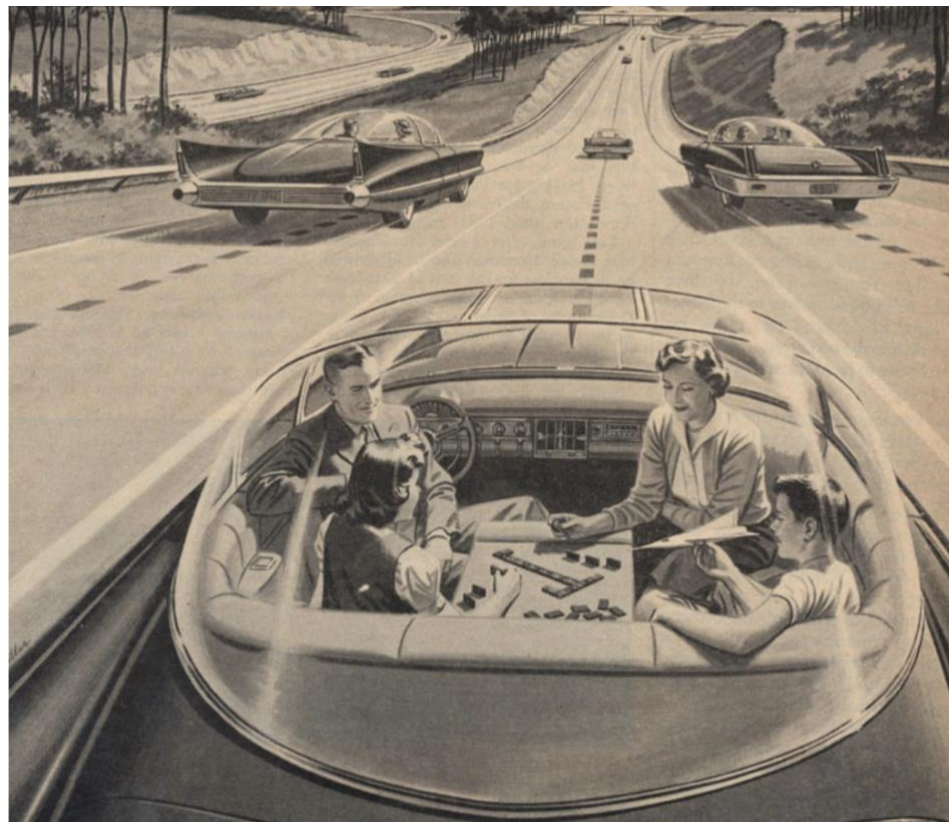


Fig. 3.5 Detailed version of the panorama—Magazine vol. 40, Nr. 5, 30. January 1956, p. 8. Image rights: copyright by the Author

(Maurer et al., 2016)

- *First Experiments and Research:* Exploration of early experimental projects and research on autonomous vehicles, such as the 1939 World's Fair "Futurama" exhibit and early attempts at self-driving cars during the mid-20th century.(Maurer et al., 2016)

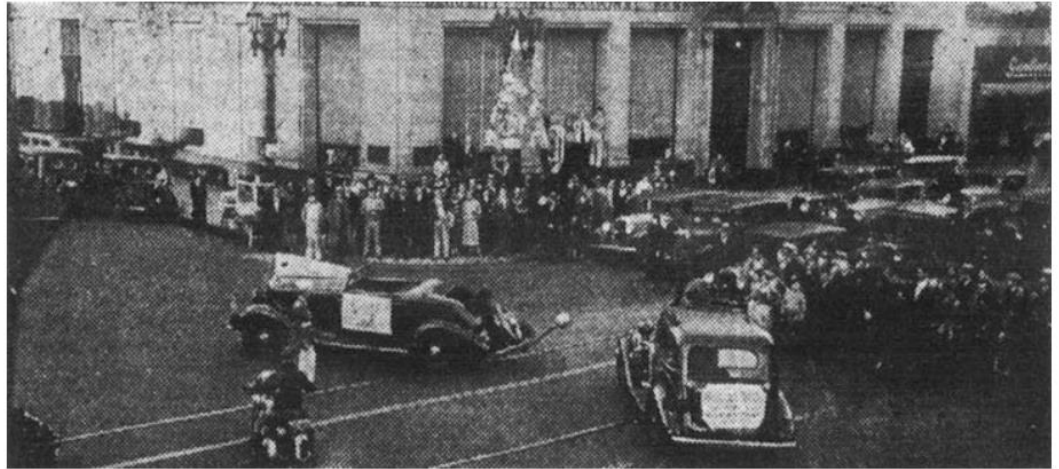


Fig. 3.2 Remote-controlled vehicle in a safety parade (USA, 1930s)—magic car to demonstrate safety, in: The Herald Statesman, 28 July, 1936, p. 1. Image rights: copyright by the Author (Maurer et al., 2016)

- *DARPA Challenges and Increased Interest:* Description of the Defense Advanced Research Projects Agency (DARPA) Grand Challenge and Urban Challenge events in the early 2000s, which significantly spurred the interest in and development of autonomous vehicle technology.(Maurer et al., 2016)

- *Major Tech and Automotive Companies' Involvement:* Overview of key technology and automotive companies entering the AV space, including Google (Waymo), Tesla, Uber, General Motors, VW Group, Daimler and Ford.(Maurer et al., 2016)

Hardware Components in AVs

- *Sensors:*

Autonomous vehicles (AVs) use a combination of sensors to perceive their environment and navigate safely. Key sensors found in AVs include:

LiDAR: Light Detection and Ranging uses laser light to measure distances and create detailed 3D maps of the surroundings.

Radar: Radio Detection and Ranging emits radio waves to detect objects, their distance, and relative speed.

Cameras: Optical cameras capture visual data to detect and recognize objects, lane markings, and traffic signs.

Ultrasonic sensors: These sensors use sound waves to detect nearby objects, particularly useful for parking and low-speed maneuvers.

Together, these sensors enable AVs to make informed decisions and respond appropriately to various traffic situations. (Maurer et al., 2016)

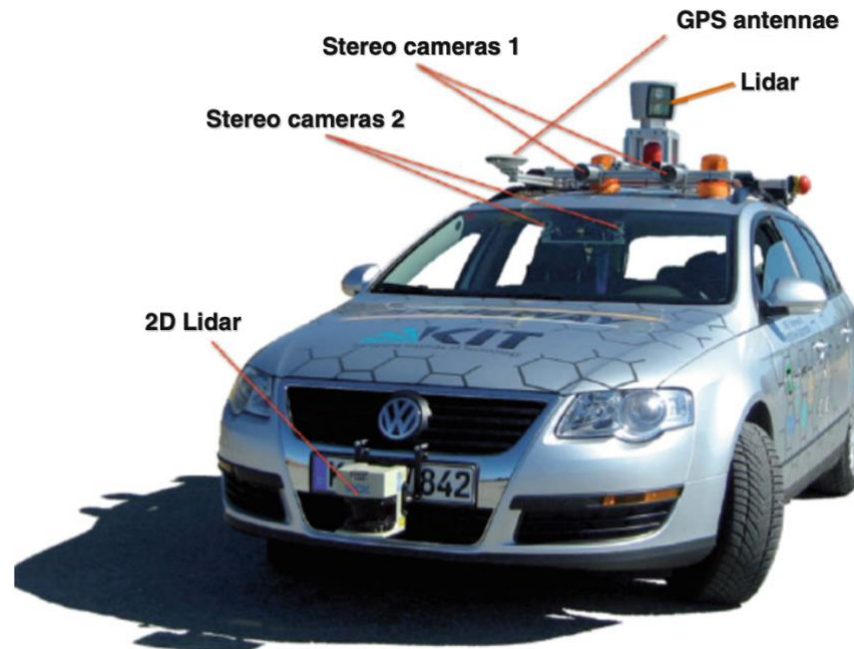


Fig. 13.3 Team anyway test vehicle, environment sensors [5]. Image rights: Springer (Maurer et al., 2016)

- Actuators and Control Systems:

These components work together to enable functions like acceleration, braking, and steering. These systems receive input from the vehicle's central processing unit (CPU), which processes sensor data and makes decisions based on algorithms and artificial intelligence.

Acceleration: Throttle actuators control the amount of air and fuel mixture that enters the engine, managing the vehicle's speed. Electronic throttle control (ETC) systems receive commands from the CPU to regulate acceleration.

Braking: Autonomous vehicles use electronically controlled braking systems, such as anti-lock braking systems (ABS) and electronic stability control (ESC). The CPU sends commands to modulate brake pressure, ensuring safe and efficient braking.

Steering: Electric power steering (EPS) systems replace traditional hydraulic systems in autonomous vehicles. The CPU sends commands to the EPS, which adjusts the steering angle based on input from sensors and the vehicle's current speed.

Together, these actuators and control systems allow the autonomous vehicle to navigate and respond to various traffic situations safely and efficiently. The seamless integration of these components is crucial for the successful operation of AVs.(Watzenig & Horn, 2017)

162

R. Schubert and M. Obst

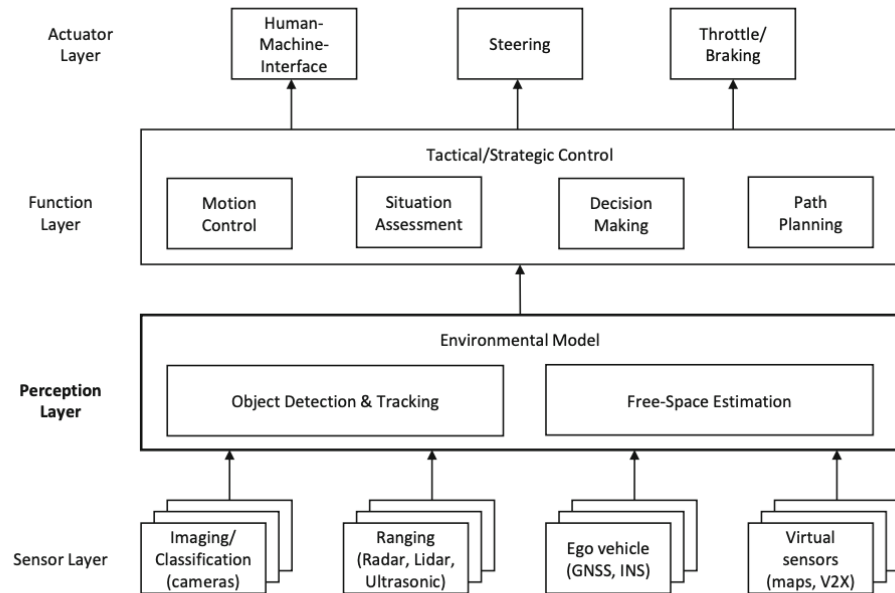


Fig. 7.1 The perception layer constitutes an interface between sensors and high-level functions

(Watzenig & Horn, 2017)

- *V2X, V2V and V2I Communication:*

V2X (Vehicle-to-Everything) communication technologies are an essential component of autonomous vehicles, enabling them to exchange information with other vehicles, infrastructure, and devices to improve safety, efficiency, and traffic flow. Two critical subsets of V2X communication are Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication.

Vehicle-to-Vehicle (V2V): V2V communication allows vehicles to exchange information directly with each other, such as their speed, position, and direction, to avoid collisions and coordinate movements. This technology can help vehicles anticipate and react to sudden changes in traffic, such as abrupt braking, lane changes, or merging vehicles, resulting in improved road safety and reduced congestion.

Vehicle-to-Infrastructure (V2I): V2I communication involves the exchange of data between vehicles and infrastructure elements, such as traffic lights, road signs, and sensors. This technology enables vehicles to receive real-time information about traffic conditions, roadworks, and potential hazards, allowing them to adjust their routes, speed, and driving strategies accordingly.

These technologies, combined with advanced sensor systems and artificial intelligence, provide autonomous vehicles with a comprehensive understanding of their environment, allowing them to make informed decisions and adapt to varying traffic situations.(Maurer et al., 2016)(Watzenig & Horn, 2017)

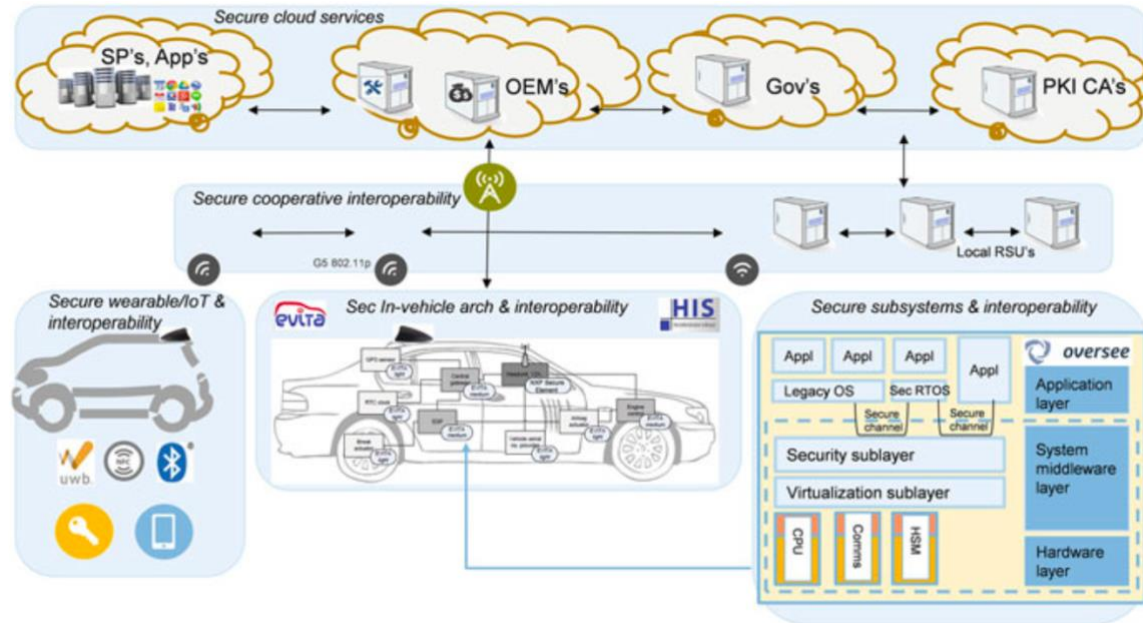


Fig. 2.1 The vehicle and its components, integrated in the Internet-of-Things, including security requirements

(Watzenig & Horn, 2017)

Software Components and Artificial Intelligence in AVs

- Localization and Mapping:

A crucial aspect of autonomous vehicle functionality is the ability to precisely determine the vehicle's position and build detailed maps of the environment. This is achieved through a combination of software systems and techniques.

Simultaneous Localization and Mapping (SLAM) is one such technique that allows the vehicle to create a map of its environment while simultaneously determining its position within the map. SLAM uses sensor data from cameras, LiDAR, radar, and other sources to construct a real-time, detailed representation of the surroundings, enabling the vehicle to navigate safely and efficiently.

Another essential component in the vehicle's positioning system is the Global Positioning System (GPS), which provides location information based on satellite signals. While GPS can offer accurate position data, it may not always be sufficient for precise navigation in complex urban environments. Therefore, autonomous vehicles typically combine GPS data with information from other sensors.

By integrating SLAM, GPS, and other sensor data, autonomous vehicles can effectively determine their position and navigate through various traffic situations, ensuring both safety and efficiency in their operations.(Maurer et al., 2016)

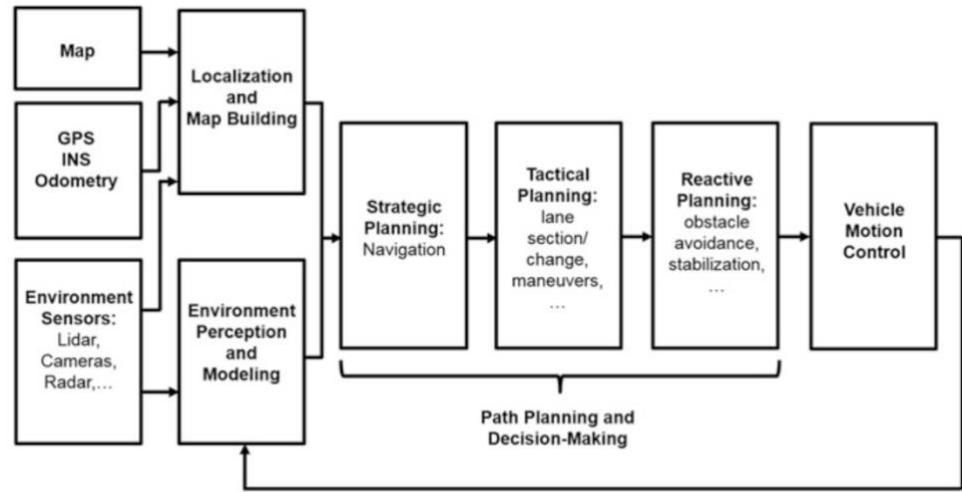


Fig. 1.3 Environment perception and modeling based on multi-sensor fusion [13]
(Watzenig & Horn, 2017)

- *Algorithms:*

Path planning algorithms determine the best route for a vehicle to follow by considering factors like traffic, road conditions, and legal constraints. Perception algorithms process sensor data to understand the vehicle's environment. Object recognition algorithms identify and track objects, such as vehicles, pedestrians, and obstacles. Control algorithms manage the vehicle's actuators to ensure smooth acceleration, braking, and steering based on the input from the central processing unit.(Watzenig & Horn, 2017)

Artificial Intelligence in Autonomous Vehicles

- *Machine Learning and Deep Learning:* Machine learning, originating from the work of McCulloch and Pitts in 1948, involves creating algorithms and models to learn from data. It encompasses various learning categories, including supervised, unsupervised, reinforcement, batch, and online learning. Techniques such as decision trees, artificial neural networks, genetic algorithms, and support vector methods are applied to address these problems. The effectiveness of machine learning methods relies on the quality, relevance, and volume of training data. Additionally, performance assessment plays a critical role in evaluating the success of these techniques, with a focus on handling noisy, inaccurate, or partial data and ensuring that the experience gained from operation is continually incorporated.(Maurer et al., 2016)

- *Neural Networks*: Artificial neural networks were initially examined for vehicle dynamic control in. They consist of interconnected neurons with weights adjusted based on training data or online. This creates an adaptive network capable of learning. The network can mimic driver reactions during a specific training phase, leading to a controller designed from the resulting model. The primary limitations of this approach include the need for training data and the inability to explain failures.(Watzenig & Horn, 2017)

- *Reinforcement Learning*: An agent collects data by performing a task to optimize it, with feedback shaping further optimization steps. Reinforcement learning faces the innovation dilemma, balancing exploration (search, risk-taking, innovation) and exploitation (refinement, efficiency, execution). The challenge is to achieve optimal execution in a partly unknown area while considering external constraints like cost, safety, and time. The learning problem is also defined by the availability and use of training data.(Maurer et al., 2016)

Infrastructure Requirements for AV Deployment

Car manufacturers worldwide are developing research prototypes at different SAE levels, but a common development roadmap is still missing. Vehicle and infrastructure research must join forces to drive innovation. Key infrastructure topics include traffic management, digital infrastructure, data backbone, cloud computing, intelligent road standards, vehicle-infrastructure interaction, IT security, privacy, and legal constraints. Investigating these areas is crucial for ensuring safe, reliable, and secure automated vehicles within an intelligent environment. Progress in legislation, liability, and insurance is also urgently needed.(Watzenig & Horn, 2017)

3.2 Ethics in Autonomous Vehicles

Overview

This chapter explores the complex challenges autonomous vehicles face during critical traffic scenarios. It delves into the decision-making processes of these vehicles when confronted with situations requiring prioritization of safety for various parties, such as passengers and pedestrians.

Ethical Dilemmas in Critical Traffic Situations

- *Trolley Problem*:

Introduced by Philippa Foot and later popularized by Judith Jarvis Thomson, is a moral dilemma that helps to highlight the differences between utilitarianism, deontology, and virtue ethics. In the problem, a person must decide whether to pull a lever and divert a runaway boxcar to save five people on the main track while killing one person on a side track, or do nothing and let the five people die. Utilitarians typically choose to pull the lever, as it maximizes overall happiness by saving more lives. A variant of the problem involves pushing a fat man in front of the boxcar to save the five people, which a deontologist would likely disapprove of, as it treats the fat man as a means to an end. The Trolley Problem and its variants serve as useful tools for exploring the reflective equilibrium process and understanding ethical perspectives.(Awad, 2017)

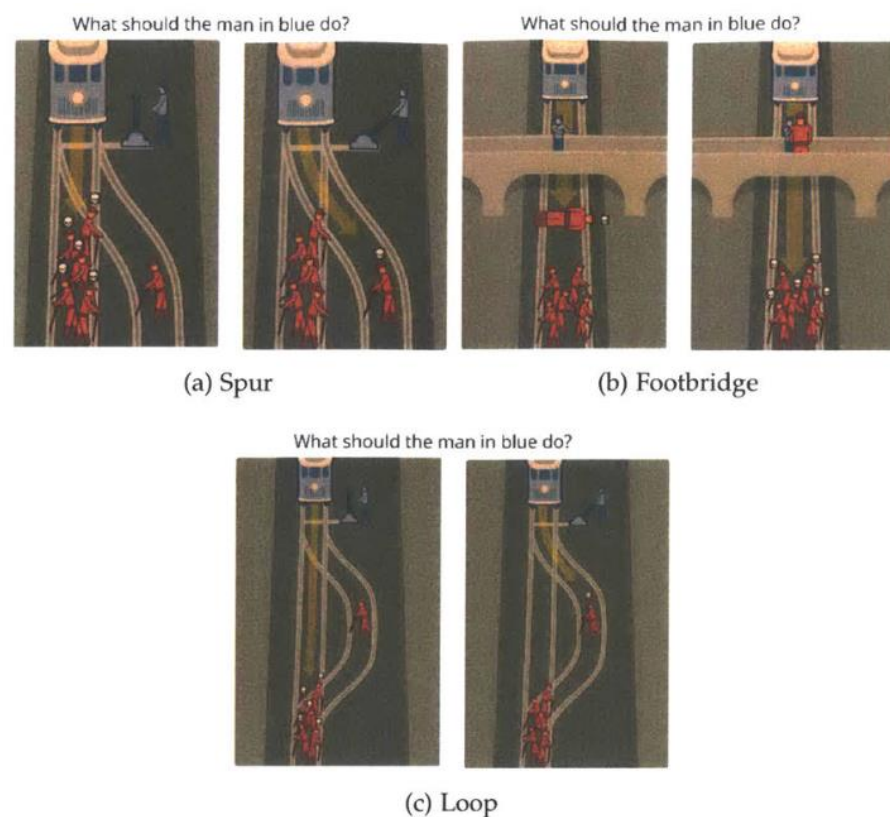


Figure 4: A visual depiction of the *Trolley Problem* – *Moral Machine* interface. (a) The original and most famous version – Spur. (b) A famous variant of the *Trolley Problem* – Footbridge. (c) Another famous variant of the *Trolley Problem* – Loop.

(Awad, 2017)

- *Life and Death Decisions*: Challenges faced by AVs when making decisions that involve prioritizing the safety of one individual or group over another, such as pedestrians versus passengers.

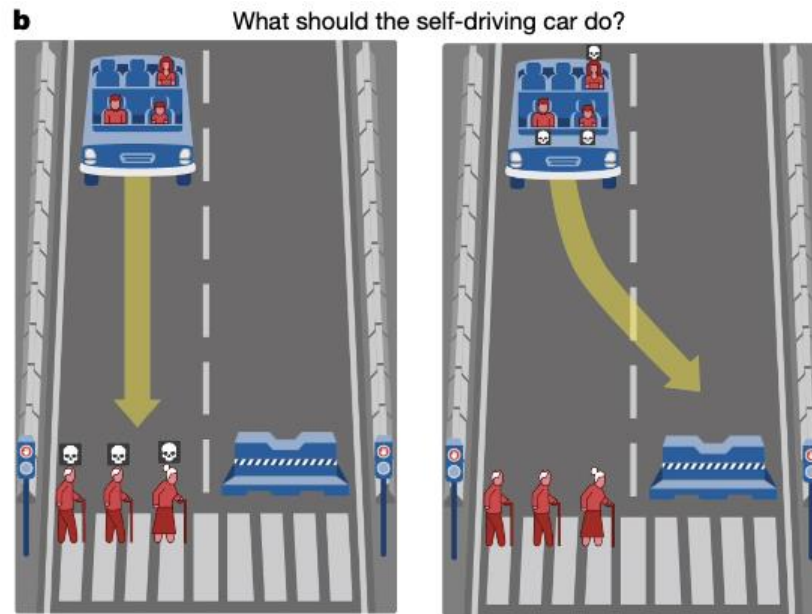


Fig. 1 | Coverage and interface. a, World map highlighting the locations of Moral Machine visitors. Each point represents a location from which at least one visitor made at least one decision ($n = 39.6$ million). The numbers of visitors or decisions from each location are not represented. **b**, Moral Machine interface. An autonomous vehicle experiences a sudden brake failure. Staying on course would result in the death of two elderly men and an elderly woman who are crossing on a 'do not cross' signal (left). Swerving would result in the death of three passengers: an adult man, an adult woman, and a boy (right).

(Awad, 2017)

3.3 Ethical Theories and Approaches

Ethical Theories and Approaches

- Normative ethics:

Also known as prescriptive ethics, consists of theories that determine the basis for judging an action's morality. It focuses on three main approaches: deontology, consequentialism, and virtue ethics. Deontology emphasizes the action and its relation to the agent's internal state, consequentialism focuses on the outcome, and virtue ethics centers on the agent's inherent virtues and moral character. While these approaches may agree on certain judgments, they do so for different reasons. Other hybrid approaches combine elements from these three main approaches.(Awad, 2017)

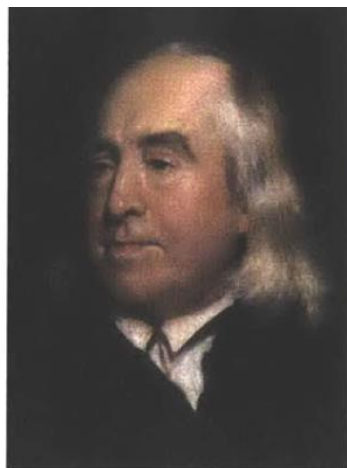
- Kantian Deontology:

A moral philosophy developed by Immanuel Kant, revolves around the concept of categorical imperatives, which are absolute moral requirements. Deontological theories focus on obligations and duties, rather than

consequences. In the given scenario, if you refuse to threaten the criminal with torture due to the belief that it should never be inflicted on any human being, you are more likely to align with deontological theories. One of the key principles of Kantian Deontology is that humans should never be treated merely as a means to an end, but always as an end themselves.(Awad, 2017)

- *Utilitarianism:*

Is the main paradigm of consequentialism, contends that an action is morally permissible if it maximizes happiness and minimizes pain for the majority. Jeremy Bentham, the father of utilitarianism, introduced methods to calculate pleasure and pain. This moral philosophy focuses on the consequences of actions and reduces morally relevant factors to outcomes, which are further reduced to a single value - utility. Despite appearing simple, classic utilitarianism underlies various complex claims about the moral rightness of an act. (Awad, 2017)



(a) Jeremy Bentham (1748-1832) [61]



(b) Immanuel Kant (1724-1804) [81]

Figure 3: Jeremy Bentham and Immanuel Kant, leading figures of *consequentialism* and *deontology*, respectively.

(Awad, 2017)

Asimov's Laws and their Relevance to Autonomous Vehicles

- *Introduction to Asimov's Laws:* Brief explanation of Isaac Asimov's Three Laws of Robotics and their origins in science fiction literature.

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given to it by human beings, except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws.

Later Asimov added a zeroth law: A robot may not harm humanity, or, by inaction, allow humanity to come to harm. (Awad, 2017)

Laws for establishing moral principles in autonomous machines have been questioned due to ambiguity, inapplicability, and misrepresentation. Embedding moral principles in robots remains unclear, with approaches like decision tree algorithms and learning from observation being debated. Decision trees provide transparency and predictability, while the learning approach handles complex, evolving human ethics. There are arguments regarding whether machines could ever be considered moral agents, focusing on their intentions, freedom, and ethical impact. Moor defined four types of ethical machines, each with increasing levels of ethical agency. (Awad, 2017)

Key Ethical Challenges

The Moral Machine survey, conducted by MIT Media Lab, gathered data from millions of respondents worldwide to study the ethical preferences of individuals in the context of autonomous vehicle decision-making in critical situations. The survey presented various scenarios where an autonomous vehicle had to decide between two outcomes, often involving the trade-off between the lives of different individuals. By analyzing the responses, the researchers were able to identify key regional differences in moral and ethical preferences.

Location-based differences in preferences are also interesting and potentially highly indicative of cultural differences. Upon aggregating responses per country, some consistent patterns that might indi-

cate broad cultural differences arise. For example, looking at Figure 15 (a), (c), (e), (g), one can see that Eastern countries, on average, appear less utilitarian, more inclined to save passengers, more inclined to avoid intervention, and more inclined to save lawful pedestrians than Western Europe and the Americas. Geographical differences on the state level are less obvious, though can be also indicative of cultural and political differences. For example, Figure 15 (b) shows that north-eastern states are more utilitarian than the rest.

Statistical analyses will be conducted to identify the decision rules that best reflect the weights and ranks that respondents give to the

various parameters manipulated in the scenarios. Additionally, further analysis will focus on how these differences can be explained by other measures collected on the country level such as GDP per capita, IQ, beliefs in Heaven and Hell, trust, and collectivism. Further, the addition of the internationalization will help better understand these cultural differences for non-English-speaking countries. (Awad, 2017)

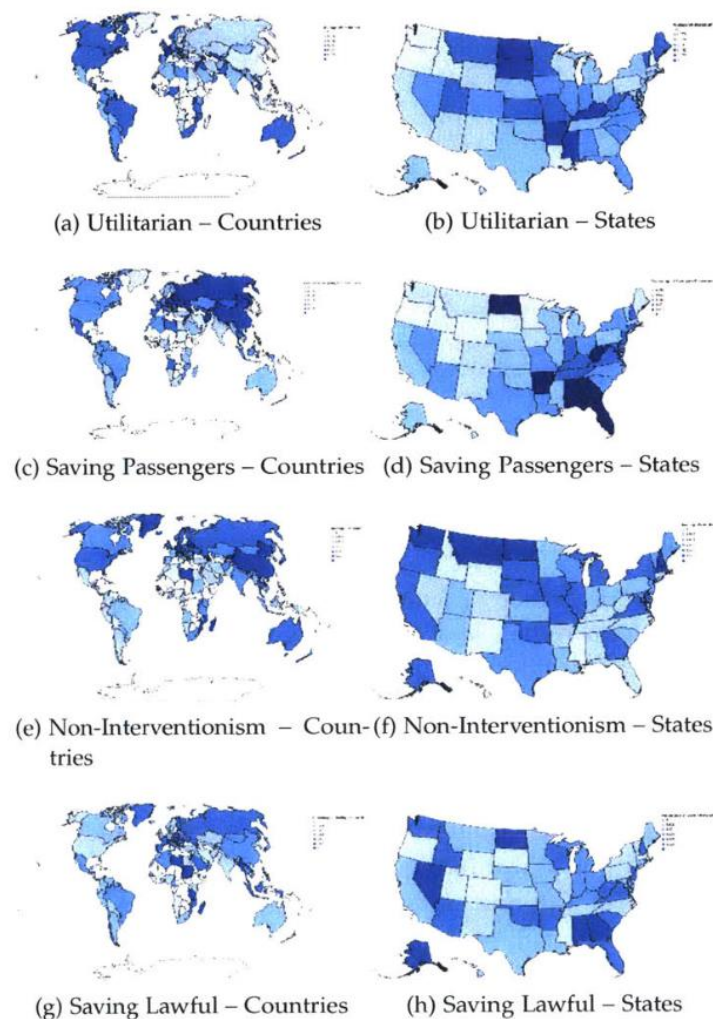


Figure 15: World map and US map highlighting (a)-(b) utilitarianism – saving more people, (c)-(d) tendency to saving passengers as opposed to pedestrians, (e)-(f) non-interventionism – tendency to leave the AV on its track and avoid swerving, and (g)-(h) saving the lawful pedestrians.

(Awad, 2017)

Pedestrians vs. Passengers: The survey found variations in how different regions valued the lives of pedestrians compared to vehicle passengers. In some cultures, there was a stronger emphasis on protecting pedestrians, even at the expense of vehicle passengers. In others, there was a more balanced consideration of the lives of both pedestrians and passengers.(Awad, 2017)

Rule-following vs. Rule-breaking: In some regions, respondents preferred saving the lives of those who were following traffic rules, such as pedestrians crossing at designated crosswalks. In contrast, other regions

showed less concern for whether individuals were following traffic rules when making their ethical decisions.(Awad, 2017)

These key regional differences highlighted by the Moral Machine survey demonstrate that ethical considerations in autonomous vehicle decision-making are not universally agreed upon. Understanding these differences is crucial for developing global guidelines and standards that address the ethical challenges posed by the widespread adoption of autonomous vehicles.(Awad, 2017)

Concepts and Principles

- *Introduction to SAE Levels of Automation:* The Society of Automotive Engineers (SAE) classification system for autonomous vehicles defines six levels of automation (Level 0 to Level 5) based on human involvement and vehicle autonomy. Level 0 represents no automation, with the driver fully in control. Level 1 introduces driver assistance features, like cruise control. Level 2 offers partial automation, where the vehicle can control steering and acceleration but requires driver monitoring. Level 3 features conditional automation, where the vehicle manages most functions, but the driver must be ready to intervene. Level 4 provides high automation, enabling the vehicle to operate independently under specific conditions. Finally, Level 5 represents full automation, with the vehicle capable of handling all driving tasks without human intervention.(Watzenig & Horn, 2017)



SAE J3016™ LEVELS OF DRIVING AUTOMATION™

Learn more here: [sae.org/standards/content/j3016_202104](https://www.sae.org/standards/content/j3016_202104)

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	SAE LEVEL 0™	SAE LEVEL 1™	SAE LEVEL 2™	SAE LEVEL 3™	SAE LEVEL 4™	SAE LEVEL 5™
What does the human in the driver's seat have to do?	You are driving whenever these driver support features are engaged – even if your feet are off the pedals and you are not steering			You are not driving when these automated driving features are engaged – even if you are seated in "the driver's seat"		
	You must constantly supervise these support features; you must steer, brake or accelerate as needed to maintain safety			When the feature requests, you must drive	These automated driving features will not require you to take over driving	

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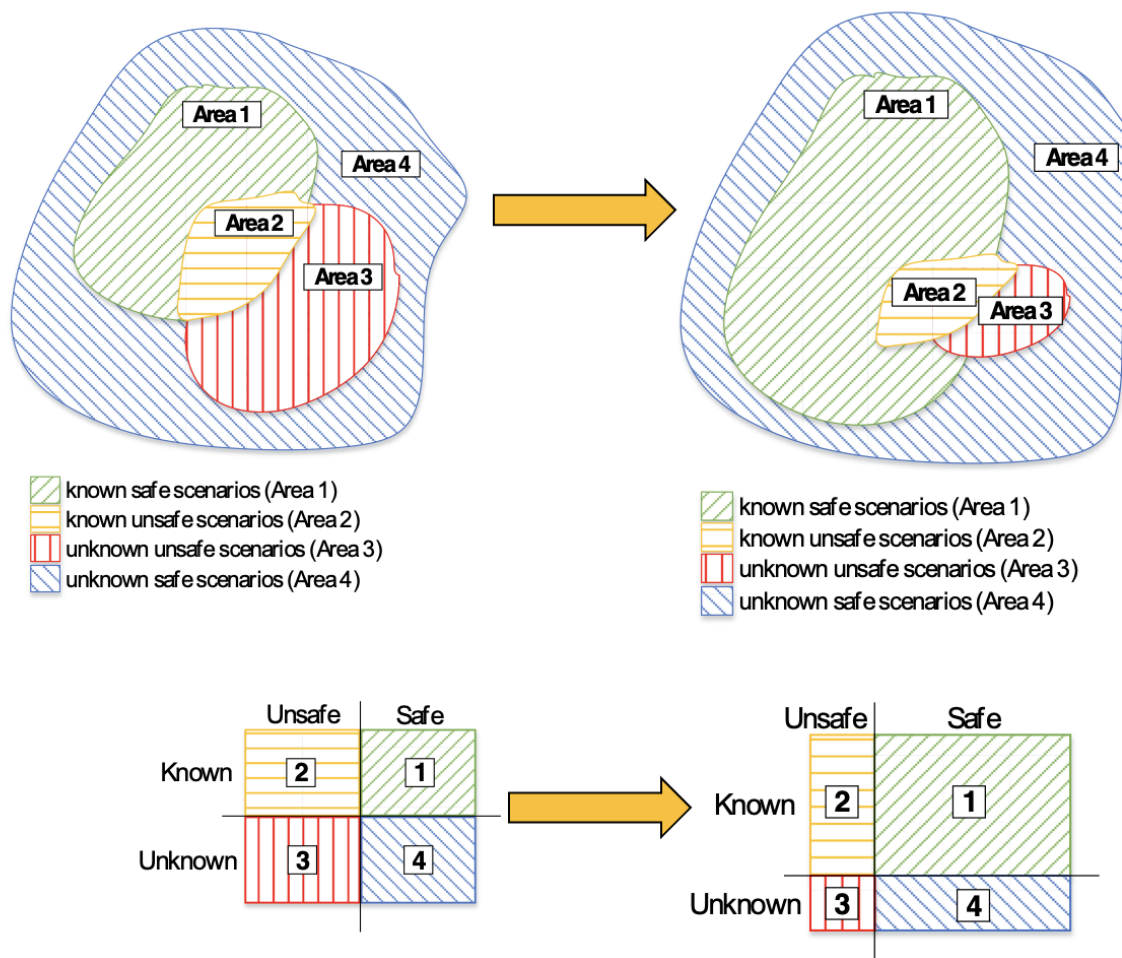
	These are driver support features			These are automated driving features		
What do these features do?	These features are limited to providing warnings and momentary assistance	These features provide steering OR brake/acceleration support to the driver	These features provide steering AND brake/acceleration support to the driver	These features can drive the vehicle under limited conditions and will not operate unless all required conditions are met	This feature can drive the vehicle under all conditions	
Example Features	• automatic emergency braking • blind spot warning • lane departure warning	• lane centering OR • adaptive cruise control	• lane centering AND • adaptive cruise control at the same time	• traffic jam chauffeur	• local driverless taxi • pedals/steering wheel may or may not be installed	• same as level 4, but feature can drive everywhere in all conditions

(SAE Levels of Driving Automation™ Refined for Clarity and International Audience, n.d.)

4 Process Maps and Risk Matrices

Overview

This chapter highlights the importance of process maps and risk matrices in the development of autonomous vehicles (AVs) and their role in enhancing safety and risk mitigation. We discuss the necessity of these tools in navigating the complexities of AV decision-making



(Walker et al., 2019)

Introduction to Process Maps and Risk Matrix

- Process maps are visual representations of the steps involved in a process, providing an overview of the sequence of events, decision points, and potential outcomes.

- In the context of autonomous vehicles and critical traffic situations, process maps can help identify potential risks and areas of concern by mapping the interactions between AVs, human-driven vehicles, pedestrians, cyclists, and traffic infrastructure.

(Ward & Chapman, 2011)

Creating Process Maps for Traffic Situations

- *Selecting traffic situations:* Choose a critical traffic situations that involve autonomous vehicles, such as intersection accidents, lane change and merging accidents, pedestrian and cyclist accidents, and rear-end collisions.
- *Identifying key steps:* Break down each traffic situation into its component steps, focusing on the interactions and decision-making processes that occur between AVs, human road users, and the traffic environment.
- *Mapping interactions and decisions:* Create a visual representation of each traffic situation, highlighting the sequence of events, decision points, and potential outcomes that may result from the actions of autonomous vehicles and other road users.

(Walker et al., 2019)

4.1 Selection of Traffic Situations for Process Mapping

Criteria for Selection

- *Relevance:* Choose traffic situations that involve autonomous vehicles and are representative of common scenarios that AVs may encounter in real-world environments.
- *Complexity:* Select situations that involve multiple decision points or interactions with other road users, as these scenarios are more likely to present challenges and risks.
- *Severity:* Prioritize traffic situations that may result in severe consequences, such as accidents involving injuries or fatalities, as these scenarios warrant greater attention in risk management efforts.
- *Diversity:* Ensure that the selected traffic situations represent a diverse range of scenarios, such as urban and rural environments, different weather conditions, and various road types, to capture a comprehensive view of the potential risks associated with AV deployment.

(Awad, 2017)

Examples of Traffic Situations for Process Mapping

- *Intersection Accidents*

AVs navigating through intersections with various traffic control mechanisms, such as stop signs, traffic lights, or roundabouts, in the presence of human-driven vehicles, pedestrians, and cyclists.

- *Lane Change and Merging Accidents*

AVs performing lane changes or merging onto highways, considering interactions with other vehicles, the need to adjust speed, and the recognition of road markings and signage.

- *Pedestrian and Cyclist Accidents*

AVs interacting with pedestrians and cyclists in various situations, such as crosswalks, shared paths, or intersections, accounting for unpredictable human behavior and the need for quick decision-making.

- *Rear-end Collisions*

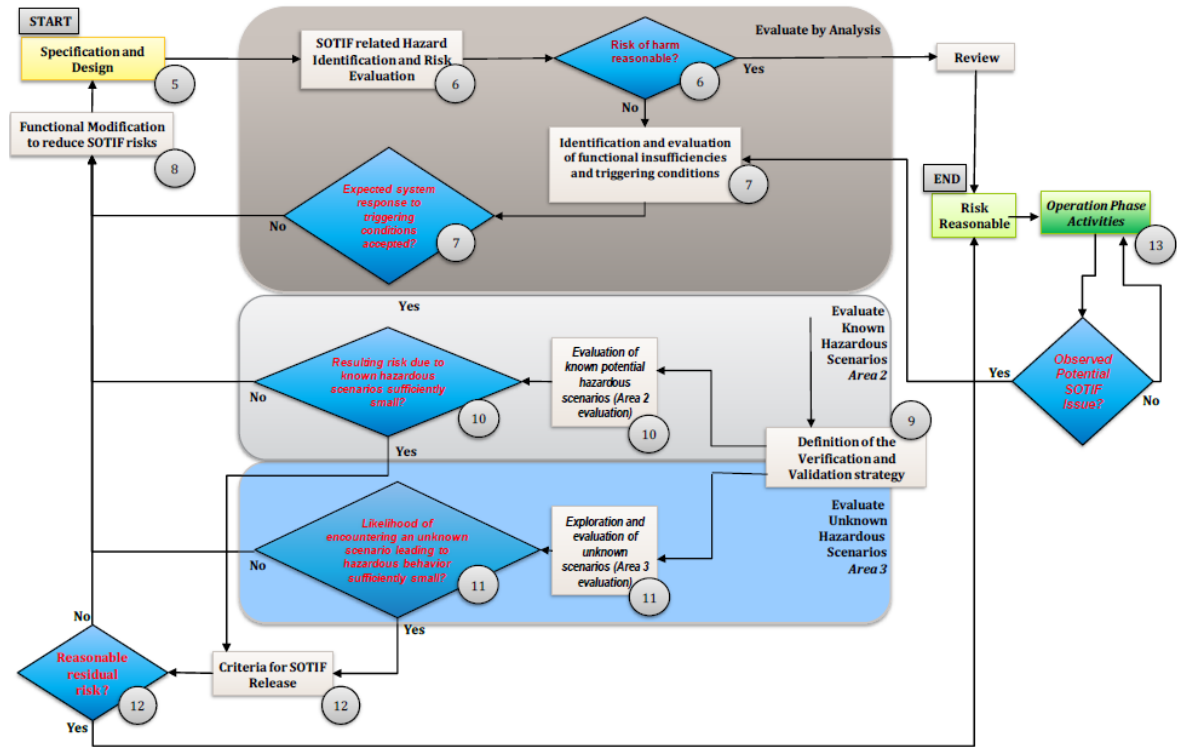
AVs reacting to sudden stops or slowdowns by other vehicles, evaluating their ability to maintain a safe following distance, respond to changing traffic conditions, and avoid rear-end collisions.

- *Low-visibility Scenarios*

AVs operating in situations with low visibility due to weather conditions, such as fog, heavy rain, or snow, which may affect the performance of sensors and decision-making systems.

By selecting and analyzing a diverse range of traffic situations, the process maps can provide a comprehensive understanding of the risks and challenges associated with autonomous vehicle deployment in critical traffic scenarios.

(Watzenig & Horn, 2017)



(Walker et al., 2019)

4.2 Detailed Process Maps for Selected Situations

Overview

This chapter presents detailed process maps for the selected critical traffic situations involving autonomous vehicles. Each process map provides a visual representation of the sequence of events, interactions, decision points, and possible outcomes associated with the specific traffic situation. The detailed process maps serve as a foundation for risk identification, analysis, and management.

4.3 Risk Matrix

Overview

In this chapter, I explore the role of risk matrices in the decision-making process of autonomous vehicles (AVs) and their importance in evaluating and mitigating potential hazards. I discuss the main elements of risk matrices, including the assessment of likelihood and severity of potential risks, and their application in various traffic scenarios involving AVs.

System behaviour	Triggering system intervention	System intervention in a critical situation (True positive)	System intervention in a NO critical situation (False positive)
	NO system intervention	NO system intervention in a critical situation (False negative)	NO system intervention in NO critical situation (True negative)
		Given critical situation	NO critical situation

Traffic situation

(Walker et al., 2019)

Introduction to Risk Matrix

- A risk matrix is a tool used to assess the likelihood and impact of risks associated with a process, providing a basis for prioritizing and managing those risks. (Anthony (Tony)Cox Jr, 2008)
- In the context of AV deployment in critical traffic situations, a risk matrix can help identify the most significant risks associated with various traffic scenarios and inform the development of risk mitigation strategies.

Developing a Risk Matrix for Traffic Situations

- Defining risk categories: Establish categories for assessing the likelihood and impact of risks associated with each traffic situation, such as low, medium, and high likelihood or impact.
- Assigning risk scores: For each traffic situation, evaluate the likelihood and impact of risks according to the predefined categories and assign a numerical score or rating to reflect their relative significance.
- Populating the matrix: Create a matrix or grid with the likelihood categories on one axis and the impact categories on the other, and place each risk in the appropriate cell based on its likelihood and impact scores.
- Prioritizing risks: Use the risk matrix to identify the most significant risks that require prioritization in terms of risk mitigation efforts, focusing on those with high likelihood and high impact.

(Anthony (Tony)Cox Jr, 2008)

Utilizing Process Maps and Risk Matrix for Risk Management

- Identifying risk mitigation strategies: Based on the insights gained from the process maps and risk matrix, develop targeted risk mitigation strategies to address the most significant risks associated with each traffic situation.
- Monitoring and updating: Continuously monitor and update the process maps and risk matrix as new information becomes available or as changes occur in the traffic environment, ensuring that risk management efforts remain relevant and effective.
- Informing policy and technology development: Use the insights gained from the process maps and risk matrix to inform the development of policies, regulations, and technological improvements aimed at minimizing the risks associated with autonomous vehicle deployment in critical traffic situations.

(Sartor & Orzes, 2019)

4.4 Creation of Risk Matrix

Overview

This chapter outlines the steps involved in creating a risk matrix for the critical traffic situations involving autonomous vehicles. The risk matrix provides a structured way of assessing and comparing the likelihood and impact of various risks, which can inform risk management and mitigation strategies.

Steps in Creating a Risk Matrix

Identify risks:

Utilize detailed process maps, along with findings from the quantitative research, to identify potential risks associated with autonomous vehicle deployment in critical traffic situations.

Create a comprehensive list of risks, including those associated with interactions, decision points, and outcomes.

Assess likelihood:

Estimate the probability or frequency of each identified risk occurring in real-world scenarios.

Factors to consider when assessing likelihood include historical data, expert opinions, and quantitative research findings.

Assess impact:

Evaluate the potential consequences or severity of each identified risk if it were to occur.

Consider factors such as the safety of road users, property damage, and the potential for delays or disruptions in traffic flow.

Assign risk ratings:

Assign a risk rating to each identified risk based on its assessed likelihood and impact.

The risk rating can be represented using a numerical scale, color codes, or descriptive terms (e.g., low, medium, high).

Plot risks on the matrix:

Plot each identified risk on the risk matrix based on its assigned risk rating.

The matrix should have axes representing likelihood and impact, with the risk ratings serving as coordinates.

Analyze and prioritize risks:

Examine the plotted risks on the risk matrix to identify trends and patterns.

Use the matrix to prioritize risks based on their relative likelihood and impact, focusing on those with the highest risk ratings for further analysis and mitigation.

Update and review the risk matrix:

Regularly review and update the risk matrix as new information becomes available, or as the understanding of the critical traffic situations involving autonomous vehicles evolves.

This iterative process ensures that the risk matrix remains a relevant and valuable tool for risk management.

By creating a risk matrix for the critical traffic situations involving autonomous vehicles, researchers can systematically assess, compare, and prioritize the identified risks. This structured approach to risk assessment provides valuable insights to inform the development of risk management

and mitigation strategies, ultimately contributing to the safe deployment of autonomous vehicles on public roads.

(Sartor & Orzes, 2019)(Anthony (Tony)Cox Jr, 2008)

4.5 Risk Assessment Criteria

Overview

- This expanded chapter delves deeper into the criteria used for assessing risks associated with the deployment of autonomous vehicles in critical traffic situations.
- The risk assessment criteria facilitate a consistent and systematic evaluation of the likelihood and impact of identified risks, aiding in informed decision-making when managing and mitigating these risks.

Likelihood Criteria

Rare:

Description: Risks that have a very low probability of occurrence in real-world scenarios.

Example: Failure of multiple redundant safety systems simultaneously in an autonomous vehicle.

Unlikely:

Description: Risks with a low probability of occurrence but may still happen under certain circumstances.

Example: An autonomous vehicle misinterpreting complex traffic signals, leading to a traffic violation.

Possible:

Description: Risks that have a moderate probability of occurrence, often arising during typical autonomous vehicle operation.

Example: A sensor malfunction causing an autonomous vehicle to momentarily deviate from its intended lane.

Likely:

Description: Risks with a high probability of occurrence, which are expected during normal operation of autonomous vehicles.

Example: An autonomous vehicle's failure to detect a small obstacle on the road, resulting in a collision.

Almost certain:

Description: Risks with an extremely high probability of occurrence, virtually guaranteed during autonomous vehicle operation.

Example: Autonomous vehicles requiring regular software updates to maintain optimal performance and safety.

(Watzenig & Horn, 2017)

Impact Criteria

Insignificant:

Description: Risks that have minimal consequences on the safety of road users, property, or the environment.

Example: A slight miscalculation in following distance causing a minor inconvenience for other road users.

Minor:

Description: Risks that have a low level of impact, potentially causing limited harm or damage.

Example: A false positive in an autonomous vehicle's obstacle detection system, leading to a sudden, unwarranted braking event.

Moderate:

Description: Risks with a noticeable impact, which can result in significant disruptions or harm.

Example: An autonomous vehicle failing to yield to a pedestrian at a crosswalk, resulting in an accident with minor injuries.

Major:

Description: Risks with a high level of impact, leading to severe harm or damage.

Example: A system failure in an autonomous vehicle causing it to veer into oncoming traffic, resulting in a multi-vehicle accident.

Catastrophic:

Description: Risks with extreme consequences, posing critical threats to safety, property, or the environment.

Example: A widespread hacking event targeting autonomous vehicles, causing numerous accidents and fatalities.

The expanded risk assessment criteria provide a more in-depth understanding of the likelihood and impact of identified risks associated with the deployment of autonomous vehicles in critical traffic situations. Utilizing these criteria ensures a consistent and structured approach to risk assessment, enabling the development of effective risk management and mitigation strategies that promote safe operation of autonomous vehicles on public roads.

(Watzenig & Horn, 2017)

4.6 Risk Matrix Development and Analysis

Overview

- This chapter elaborates on the development and analysis of a risk matrix for evaluating risks associated with autonomous vehicles in critical traffic situations.
- The risk matrix serves as a visual tool for understanding, comparing, and prioritizing risks, guiding risk management and mitigation efforts.

Developing the Risk Matrix

Compile identified risks:

Consolidate the list of potential risks identified from the process maps, quantitative research, and literature review.

Ensure the list is comprehensive and covers all relevant aspects of autonomous vehicle deployment in critical traffic situations.

Apply risk assessment criteria:

Use the likelihood and impact criteria outlined in to assess each identified risk.

Assign a likelihood and impact score to each risk, providing a clear basis for comparison and prioritization.

Plot risks on the matrix:

Create a matrix with axes representing likelihood and impact, divided into corresponding categories (e.g., Rare, Unlikely, Possible, Likely, Almost certain for likelihood; Insignificant, Minor, Moderate, Major, Catastrophic for impact).

Plot each risk on the matrix based on its assigned likelihood and impact scores, indicating its position relative to other risks.

(Anthony (Tony)Cox Jr, 2008)

Analyzing the Risk Matrix

Identify patterns and trends:

Examine the distribution of plotted risks on the matrix to identify any patterns or trends that may emerge.

For instance, a concentration of risks in the high-impact categories may indicate the need for more robust safety measures or further research.

Prioritize risks:

Use the risk matrix to prioritize risks based on their relative likelihood and impact.

Focus on risks in the higher likelihood and impact categories, as these pose the greatest threats to the safe deployment of autonomous vehicles in critical traffic situations.

Determine risk management strategies:

Based on the prioritized risks, develop appropriate risk management and mitigation strategies.

Consider a combination of preventive measures, such as improved system design and testing, and reactive measures, like incident response plans and fail-safe mechanisms.

Monitor and update the risk matrix:

Regularly review and update the risk matrix to reflect new information, changes in autonomous vehicle technology, or evolving traffic situations.

This ongoing process ensures the risk matrix remains a relevant and valuable tool for informed decision-making in risk management.

By developing and analyzing a risk matrix, researchers can systematically assess, compare, and prioritize the risks associated with autonomous vehicle deployment in critical traffic situations. This visual tool enables a better understanding of potential threats and their relative importance, guiding the development of effective risk management and mitigation strategies that contribute to the safe operation of autonomous vehicles on public roads.

(Anthony (Tony)Cox Jr, 2008)

5 Quantitative Research

5.1 Survey Design

Overview

This chapter focuses on the design of the survey used to collect data from respondents regarding their attitudes towards the risks associated with autonomous vehicle deployment in critical traffic situations.

It covers the development of the survey instrument, question types, response options, and the pretesting process to ensure a valid and reliable data collection tool.

Development of the Survey Instrument

Review of existing instruments: I examined previously published surveys and questionnaires related to autonomous vehicles and risk perception, identifying relevant questions and response options that can be adapted for the current study.

Development of new questions: Based on the research questions and the literature review, I created new questions that address the specific aspects of autonomous vehicle risks in critical traffic situations.

Pilot testing: I conducted a pilot test of the survey instrument with a small sample of respondents to identify any issues or difficulties with the questions, response options, or survey administration procedures.

Question Types

I chose to use 3 demographic, 2 closed-ended and 3 Likert scale questions for the research survey due to several reasons. First and foremost, these question types are straightforward for respondents to answer, simplifying the data collection process and enabling efficient gathering of responses from a larger number of participants. Moreover, closed-ended and Likert scale questions ensure consistency across respondents, as everyone is provided with the same set of answer choices, allowing for more accurate comparisons and reducing potential biases. Lastly, these types of questions generate

quantifiable data, which not only streamlines the analysis process but also facilitates the application of various statistical methods, enabling a comprehensive examination of the research objectives.

Questions and Response Options

1. You are:
 - a. Younger than 30
 - b. Older than 30

2. Which of the following best describes your background?
 - a. AI/Automotive Engineer
 - b. Other Engineering Professional
 - c. Student in AI/Automotive Engineering or related field
 - d. Student in a non-related field
 - e. General public with no specific background in AI/Automotive Engineering

3. You are an active driver
 - a. Yes
 - b. No

4. On a scale of 1 to 5, how confident are you in the ability of autonomous vehicles to safely navigate through complex traffic situations?
 - a. 1-5

5. If an autonomous vehicle encounters an unavoidable accident, which factor should have the highest priority in determining the vehicle's response?
 - a. Minimizing the OVERALL number of injuries
 - b. Protecting PASSENGERS in the vehicle
 - c. Protecting vulnerable road users, such as PEDESTRIANS and cyclists
 - d. Minimizing property damage
 - e. Adhering to traffic rules and regulations (e.g., collide with pedestrians who are crossing illegally)

6. How confident are you in the ability of current AI technology to make morally appropriate decisions in critical traffic situations compared to human drivers?
 - a. 1-5

7. In a critical traffic situation where an autonomous vehicle becomes aware that it cannot handle the situation effectively, which of the following expected behaviors do you think is the most appropriate for the vehicle to prioritize?
 - a. Alerting the human driver immediately and providing sufficient time for them to take over control, if available
 - b. Attempting to slow down and find a safe location to stop while waiting for human intervention
 - c. Activating emergency lights and audible warnings to alert surrounding drivers and pedestrians
 - d. Continuing to operate autonomously while continuously assessing the situation and updating its strategy
 - e. Connecting to external sources (e.g., infrastructure, nearby vehicles) for real-time assistance)

8. How confident are you in the ability of current AI technology to make quick and effective decisions in critical traffic situations compared to human drivers?
 - a. 1-5

Pretesting and Refinement

I conducted cognitive pretesting of the survey by first selecting a small group of individuals representing the target population. These individuals were asked to complete the survey while sharing their thoughts and any difficulties they encountered. After the pretest, I collected their feedback, analyzed potential issues, and made necessary adjustments to the survey design to ensure clarity, comprehension, and relevance before launching the final version.

I conducted usability testing for the survey by inviting a small group of participants, representative of the target population, to complete the survey online. This process aimed to identify any navigational, technical, or design issues that could potentially hinder the user experience. Upon collecting

feedback from the participants, I refined the survey by addressing the identified issues, making adjustments to the layout, question format, and overall presentation to ensure a smooth and user-friendly experience for the respondents.

5.2 Sampling Strategy and Population

Overview

This section outlines the sampling strategy and target population for the study, ensuring that the collected data is representative and generalizable to the broader group of interest.

It provides an in-depth description of the target population and explains the rationale behind the chosen sampling method.

Target Population

The target population for this study includes mainly general public, students and AI/AV professionals. These groups were selected for the following reasons:

1. **Relevance to Autonomous Vehicles:** Engineering students and professionals, particularly those in AI, automotive, or related fields, are likely to have a keen interest in autonomous vehicle technology. Their knowledge and understanding of the underlying systems, challenges, and advancements make their opinions and perspectives valuable for this study.
2. **Age-based Diversity:** By including respondents from both below and above 30, the study captures the views of different age groups, which may have varying attitudes towards autonomous vehicles. Younger respondents, who have grown up with rapid technological advancements, may be more comfortable with and accepting of new technologies like autonomous vehicles. In contrast, older respondents may be more cautious, having experienced the evolution of transportation technologies throughout their lives.
3. **Future Decision-Makers:** Engineering students represent the future workforce in the automotive and technology industries. Understanding their perspectives on the ethical and risk management aspects of autonomous vehicles can provide insights into how future professionals may approach these issues as they enter

the workforce and contribute to the development and deployment of AVs.

4. **Practical Experience:** Professionals aged above 30 are more likely to have practical experience in the industry, which can offer valuable insights into the real-world challenges and considerations associated with the development and deployment of autonomous vehicles.
5. **Informed Opinions:** Both engineering students and professionals have the technical knowledge to form informed opinions on the ethical and risk management aspects of autonomous vehicles, making their perspectives particularly relevant to the thesis.

By targeting these specific groups, the study aims to gather a diverse range of opinions and insights, enabling a comprehensive understanding of the critical traffic situations and risk management strategies associated with autonomous vehicles.

Sampling Strategy

Choosing a random sampling method for this study is justified for the following reasons:

1. **Representativeness:** Random sampling helps ensure that the sample is representative of the target population, as each individual has an equal chance of being selected. This minimizes potential biases and leads to more generalizable results.
2. **Simplicity:** Random sampling is relatively easy to implement, making it a practical choice for researchers with limited resources or time constraints.
3. **Reduced Bias:** Random sampling reduces the risk of selection bias, as individuals are chosen randomly and not based on any specific characteristics or preferences of the researcher.

Sample Size

In this study, I am aiming to gather a diverse range of opinions and insights on autonomous vehicles and risk management from engineering students and professionals. Since the focus is on capturing diverse perspectives rather than generalizing the findings to a larger population, a relatively smaller sample size can be justified.

A smaller sample size, in this case, offers several advantages:

1. **Feasibility:** With limited resources and time constraints, a smaller sample size is more practical to manage and analyze, ensuring that the study can be completed within the given time frame.
2. **In-depth Analysis:** A smaller sample size allows for a more detailed examination of the responses, which may provide valuable insights into the attitudes and opinions of the participants.
3. **Cost-efficiency:** A smaller sample size reduces the cost and effort involved in data collection, analysis, and interpretation, making the study more efficient.

Taking these factors into account, a sample size that balances the need for diverse perspectives and the practical constraints of the study is selected. By choosing this sample size, the study aims to achieve meaningful results while maximizing the efficiency and feasibility of the research process.

5.3 Statistical Analysis and Software Tools

Overview

This chapter discusses the statistical analysis methods and software tools used to process and analyze the collected data on respondents' attitudes towards the risks associated with autonomous vehicle deployment in critical traffic situations.

It provides an overview of the selected statistical techniques, their suitability for addressing the research questions, and the software tools used to conduct the analyses.

Data Preparation

Data cleaning is a crucial step in the data analysis process, as it ensures that the collected data is accurate, consistent, and suitable for further analysis. The process of cleaning and preparing the data involves several steps:

1. **Inspect the data:** Begin by reviewing the collected data to identify any errors, missing values, or inconsistencies that may affect the accuracy and reliability of the analysis.
2. **Remove duplicate entries:** Check for and eliminate any duplicate records that might have been inadvertently included in the dataset.

3. Address missing values: Identify any missing data points and determine the best approach for handling them, which may involve filling in the missing values with an appropriate estimate, interpolating the data, or excluding the records with missing values from the analysis.
4. Correct data entry errors: Identify and fix any data entry errors, such as typos, mislabeling, or incorrect units, to ensure that the data is accurate and consistent.

Statistical Techniques

Descriptive statistics: Use of descriptive statistical techniques to summarize and describe the main features of the dataset, including means, medians, etc.

Inferential statistics: Use of inferential statistical methods, such as t-tests.

Correlational analysis: Conduct correlational analyses, to investigate relationships between variables and identify potential associations or trends.

Software Tools

I am utilizing software tools for inferential statistics, namely Microsoft Excel and Google Forms. Microsoft Excel is a powerful spreadsheet software that allows for the efficient organization, analysis, and visualization of data. Its built-in statistical functions and features make it a suitable tool for conducting inferential statistical analysis. Google Forms is a versatile online survey tool that enables the collection of responses and seamless integration with Google Sheets, which can be easily imported into Microsoft Excel for further analysis. These tools, combined, will provide a robust and accessible platform for conducting our research and analyzing the results.

The selection of Microsoft Excel and Google Forms as software tools for inferential statistics in this thesis is justified based on their accessibility, functionality, and ease of use. Microsoft Excel is widely available and familiar to many, making it a practical choice for data analysis. Its extensive library of built-in statistical functions and features enables users to perform a range of inferential statistical tests with relative ease.

Google Forms is a user-friendly and versatile online survey tool that simplifies the process of collecting data. Its seamless integration with Google Sheets allows for efficient organization and management of collected data, which can be readily imported into Microsoft Excel for analysis. The compatibility between Google Forms and Microsoft Excel ensures a smooth

and streamlined workflow, minimizing the learning curve and potential technical barriers for researchers.

Overall, the combination of Microsoft Excel and Google Forms provides a robust and accessible platform for conducting research and analyzing data in this thesis, making them well-suited for the intended purpose.

To implement Microsoft Excel and Google Forms for inferential statistics in this thesis, follow these steps:

1. Design the survey using Google Forms, keeping questions clear and relevant.
2. Collect responses from the target audience.
3. Export data to Google Sheets, then import into Microsoft Excel.
4. Clean and organize data in Excel, grouping and formatting as needed.
5. Perform inferential statistical tests using Excel's built-in functions.
6. Create visualizations to interpret results and communicate findings.
7. Analyze results, draw conclusions, and provide recommendations based on the findings.

5.4 Interpretation and Implications of Results

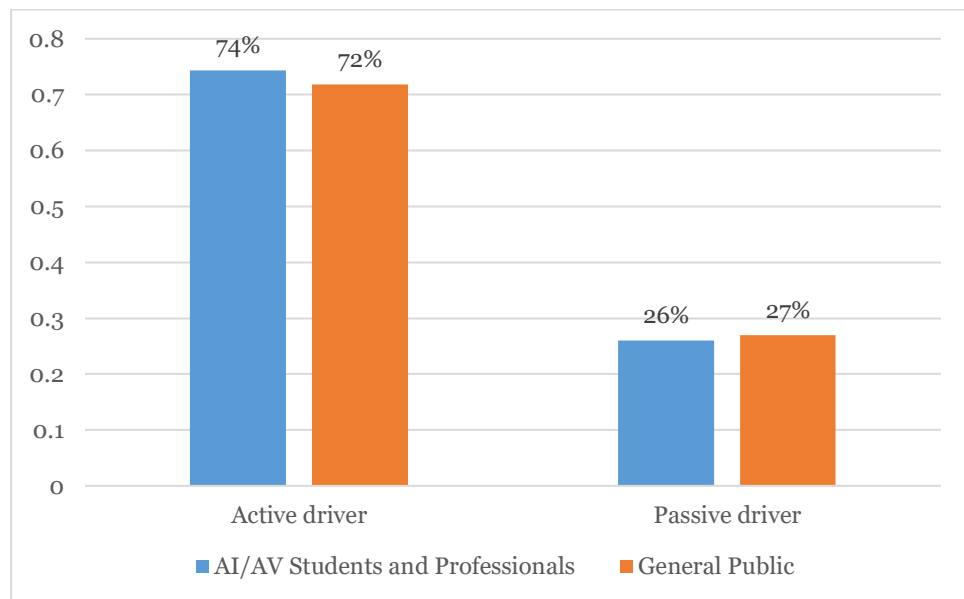
Overview

The results overview includes 117 respondents, with 39 AV/AI professionals and students and 78 individuals from the general public. A significant 89 percent of the total respondents were under 30 years old. The diverse participant pool contributes valuable insights from both field experts and the wider community, especially among younger age groups. The information obtained helps to better understand opinions and concerns related to autonomous vehicles and their ethical decision-making processes.

Charts

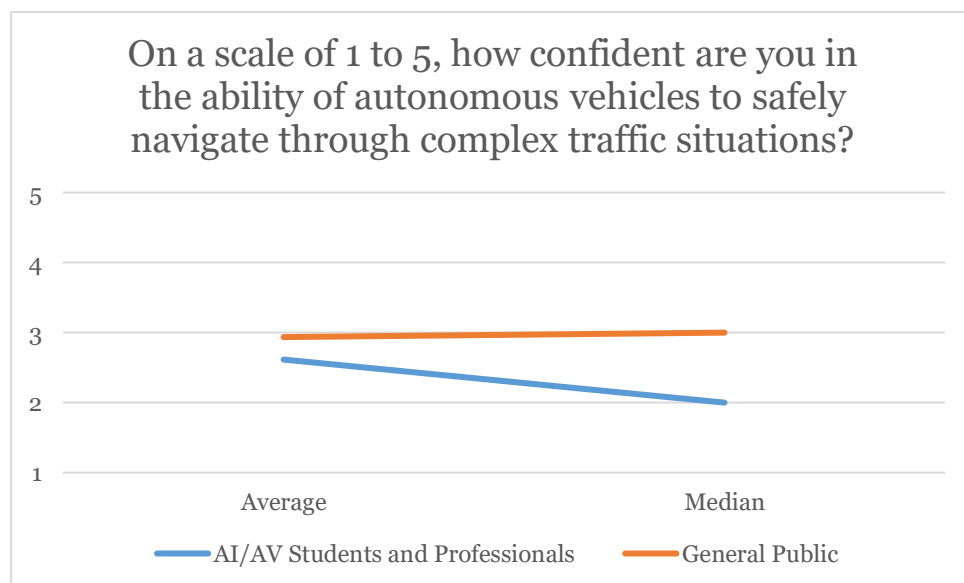
- The survey results show a striking similarity between AI/AV professionals and the general public in terms of active drivers, with 74%

for AI/AV engineers and 72% for the general public. This interesting observation indicates that both groups have comparable levels of driving exposure, potentially enabling a more balanced assessment of the survey.

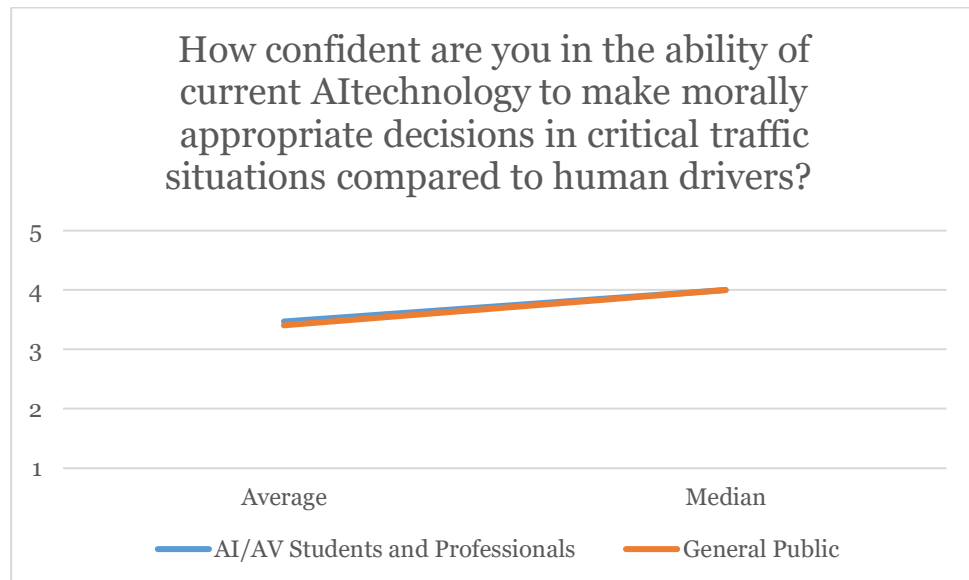


Lickert scale results

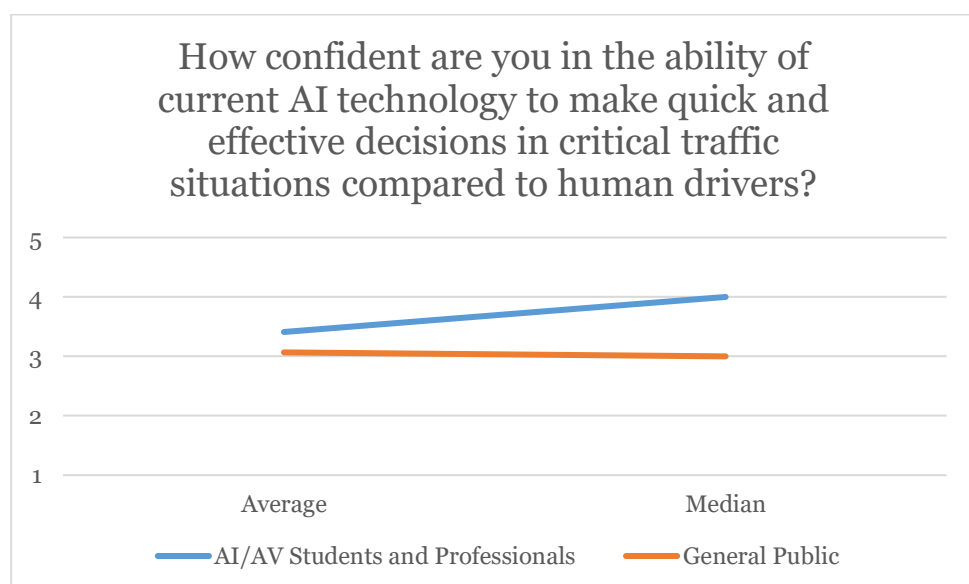
- AI/AV experts and the general public have a moderate level of confidence in the ability of autonomous vehicles to safely navigate through complex traffic situations. The mean confidence for AI/AV professionals falls between 2.5 and 3, indicating some degree of uncertainty. Interestingly, the median confidence level for AI/AV professionals is 2, while it is 3 for the general public. This suggests that the general public may have a slightly higher level of trust in autonomous vehicles' capabilities to handle complex traffic situations compared to AI/AV experts.



- The survey results reveal that both AI/AV students/professionals and the general public share similar levels of confidence in the ability of current AI technology to make morally appropriate decisions in critical traffic situations compared to human drivers. With both the median and mean being nearly the same for both groups, it suggests a consensus in the perception of AI's capabilities.

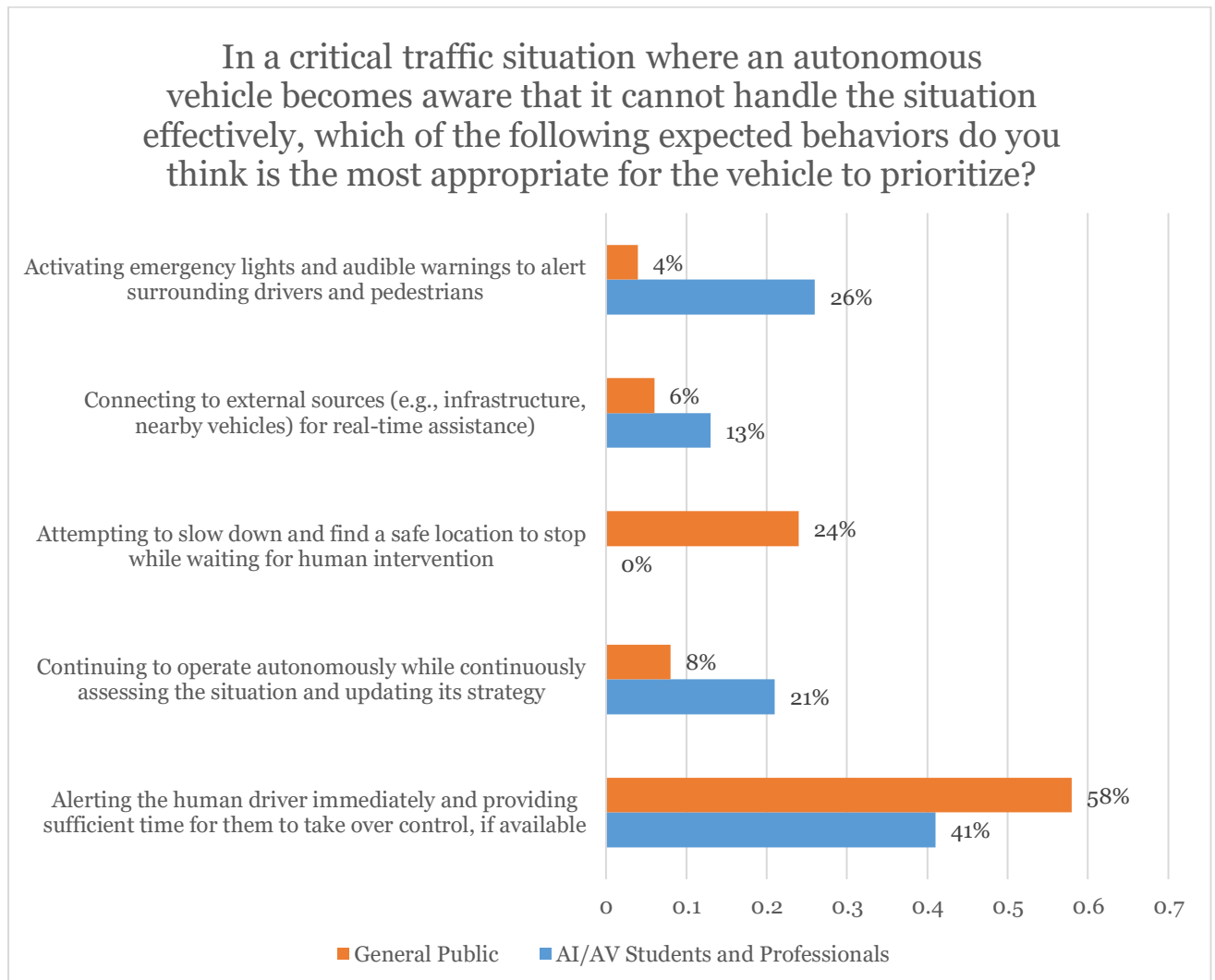


- Both experts and the general public have a moderate level of confidence in the ability of current AI technology to make quick and effective decisions in critical traffic situations compared to human drivers. Both groups reported an mean confidence score of 3 to 3.5. However, the median confidence level for AI/AV experts is 4, while it is 3 for the general public. This indicates that AI/AV experts might have a slightly higher level of trust in AI's time critical decision-making capabilities in critical traffic situations compared to the general public.



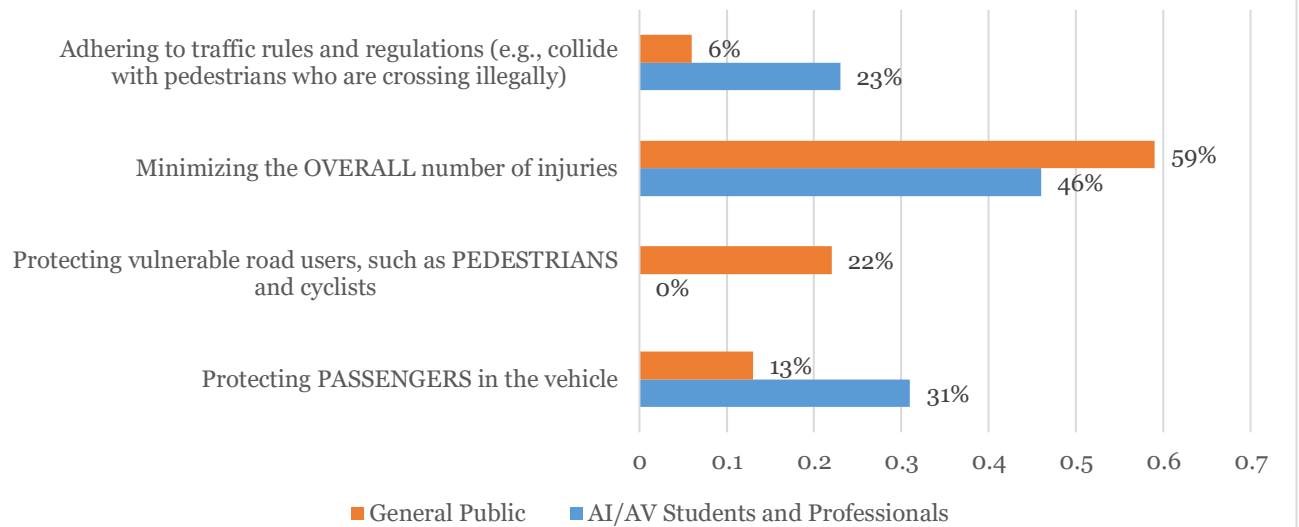
Close-ended results

- The survey results for the question regarding the most appropriate behavior for an autonomous vehicle (AV) in a critical traffic situation show some distinct preferences among AI/AV experts and the general public. These results indicate that both groups lean towards minimizing the overall number of injuries in critical traffic situations, while there are notable differences in other priorities. AI/AV experts seem to value passenger protection and adherence to traffic rules more than the general public, who show a greater concern for vulnerable road users.



- The survey results for the question regarding the most appropriate behavior for an autonomous vehicle (AV) in a critical traffic situation show some distinct preferences among AI/AV experts and the general public. These results indicate that both groups lean towards minimizing the overall number of injuries in critical traffic situations, while there are notable differences in other priorities. AI/AV experts seem to value passenger protection and adherence to traffic rules more than the general public, who show a greater concern for vulnerable road users.

If an autonomous vehicle encounters an unavoidable accident, which factor should have the highest priority in determining the vehicle's response?



6 Evaluation and Conclusion

6.1 Theoretical and Practical Implications

The theoretical and practical implications of these survey results are as follows:

Theoretical Implications:

Both AI/AV experts and the general public have similar confidence levels in the ability of AI technology to make morally appropriate decisions and navigate complex traffic situations. This highlights the need for further exploration of trust and acceptance factors in AV technology adoption.

The differences in preferences between AI/AV experts and the general public suggest that ethical frameworks and decision-making algorithms for AVs should take into account the concerns and expectations of diverse stakeholders.

The diverse range of responses indicates that there is no universally agreed-upon solution for handling critical situations, emphasizing the importance of continued research on ethical and decision-making theories in the context of AVs.

Practical Implications:

AV developers should consider incorporating features that allow human drivers to take over control in critical situations, as it emerged as a top priority for both AI/AV experts and the general public.

Education and communication campaigns can be designed to address the concerns and expectations of the general public, bridging the gap between AI/AV experts' knowledge and public perception.

Policymakers and regulators should consider the opinions of various stakeholders while developing guidelines for AV behavior in critical traffic situations, striking a balance between public concerns and technological advancements.

Collaboration between AI/AV experts, policymakers, and the general public is crucial to ensure that AV technology is both ethically sound and publicly acceptable.

These implications underline the importance of considering diverse perspectives in the development, regulation, and deployment of AV technology, as well as the need for further research and collaboration to address the complex ethical challenges that autonomous vehicles present.

6.2 Discussion of Findings and Conclusions

Overview

In this chapter, I discuss the findings of the study and draw conclusions based on the survey results. My analysis has provided insights into the opinions of AI/AV experts and the general public regarding the ethical decision-making and safety capabilities of autonomous vehicles in critical traffic situations.

Confidence in Autonomous Vehicle Decision-making

The survey results showed that both AI/AV experts and the general public have similar confidence levels in the ability of AI technology to make morally appropriate decisions and navigate complex traffic situations. This finding is interesting, as it suggests that even though AI/AV experts have more in-depth knowledge about the technology, their confidence levels are not significantly higher than those of the general public. This could indicate the need for further research on trust and acceptance factors in AV technology adoption.

Prioritization of Safety and Ethical Decision-making

The results revealed differences in the preferences of AI/AV experts and the general public when it comes to the expected behaviors of autonomous vehicles in critical traffic situations. For example, a higher percentage of AI/AV experts prioritized minimizing the overall number of injuries, while the general public showed a stronger preference for protecting vulnerable road users.

It is worth noting that these findings build on the results of MIT's Moral Machine experiment, which demonstrated significant variations in ethical preferences across different cultures. This further highlights the importance of considering diverse cultural perspectives when developing ethical guidelines for autonomous vehicles.(Awad et al., 2018)

These findings emphasize the importance of incorporating diverse stakeholder perspectives into the development of ethical frameworks and decision-making algorithms for AVs. It is crucial to ensure that AV technology addresses the concerns and expectations of both experts and the general public, as well as various cultural backgrounds, to promote trust and acceptance.

Responses to Unavoidable Accidents

In the case of unavoidable accidents, the survey results highlighted the importance of alerting the human driver immediately and providing sufficient time for them to take over control, if available. This preference was shared by both AI/AV experts and the general public, indicating the need for AV developers to consider incorporating such features in their designs.

Conclusion

My study has provided insights into the opinions of AI/AV experts and the general public regarding the ethical decision-making and safety capabilities of autonomous vehicles in critical traffic situations. The findings emphasize the importance of considering diverse perspectives in the development, regulation, and deployment of AV technology, as well as the need for further research and collaboration to address the complex ethical challenges that autonomous vehicles present.

6.3 Limitations of the Study

- *Limited sample size:* The study may have a limited sample size, which reduces the statistical power of the analysis and the ability to make reliable generalizations about the broader population. A larger sample size would improve the robustness of the findings and increase the confidence in the conclusions drawn.
-
- *Generalizability:* The findings of the thesis may not be applicable to other populations, settings, or contexts outside the specific scope of the research. The results could be influenced by factors unique to the sample or the environment in which the study was conducted, limiting the external validity of the conclusions.
-
- *Methodological limitations:* The research design, particularly the use of survey questions, may have limitations that affect the validity or reliability of the study's findings. Survey questions may be subject to interpretation, leading to potential biases in the responses. Additionally, participants may be influenced by social desirability bias or other factors that could affect the accuracy of their responses.
-
- *Survey design:* The construction and wording of survey questions may impact the responses and the overall findings of the study. Ambiguity or leading questions could influence participants' answers and potentially introduce bias. Furthermore, the response options provided in the survey

may not adequately capture the full range of opinions or experiences of the respondents.

6.4 Recommendations for Future Research

Overview

Based on the findings and limitations of the current study, I suggest the following recommendations for future research:

Recommendations

- *Larger and more diverse sample:* To improve the generalizability of the findings, future research should aim to include a larger and more diverse sample of respondents. This could involve recruiting participants from different cultural backgrounds, professions, and age groups to better understand the varied perspectives on autonomous vehicles and their ethical implications.
- *Longitudinal studies:* Conducting longitudinal research could help to examine how attitudes towards autonomous vehicles and their ethical decision-making processes change over time. This would provide valuable insights into the evolution of public opinion as the technology advances and becomes more widespread.
- *Experimental studies:* To complement the survey-based approach, future research could employ experimental designs to explore the ethical decision-making of autonomous vehicles in controlled environments. This would help to provide more objective and direct evidence on the impact of different ethical principles and decision-making algorithms on real-life traffic situations.
- *Qualitative research:* Conducting in-depth interviews or focus groups with stakeholders, including AI/AV experts, policymakers, and the general public, could provide a more nuanced understanding of their perspectives and concerns regarding the ethical implications of autonomous vehicles. This would help to identify potential areas of consensus and disagreement, informing the development of more effective and acceptable ethical frameworks.
- *Comparative studies:* Future research could compare the ethical decision-making of autonomous vehicles across different cultural and regulatory contexts. This would help to identify potential cross-cultural variations in attitudes towards the technology and inform the development of culturally-sensitive ethical guidelines.

- *Evaluation of ethical frameworks and guidelines:* As ethical frameworks and guidelines for autonomous vehicles continue to evolve, it will be essential to assess their effectiveness in guiding the technology's development and implementation. Future research should aim to evaluate the impact of these guidelines on the safety, efficiency, and public acceptance of autonomous vehicles, providing evidence-based recommendations for further improvements.

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Note: The end of the list of references should include the end of the section to allow for various types of numbers to be used in the main thesis text and appendices.